20 Years of Local Outdoor Recreation and Its Value to Americans

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Abstract

The value of outdoor recreation regularly justifies public spending on "the great outdoors." Yet 95 percent of recreation trips require an hour or less of round-trip travel. We develop an econometric approach that uses two decades of the American Time Use Survey (2003–2023) to produce repeated national estimates of the value of local recreation. A lower-bound estimate of willingness to pay per trip remained stable, between $5 and $6 (2023 USD), while national annual willingness to pay rose 63 percent, from $57.0 to $93.2 billion, driven by rising participation and population growth. Short trips to play with pets or children, walk, or socialize account for about 80 percent of outdoor recreation trips, yet these high-frequency activities are largely absent from the valuation literature. A relative price change framework, combined with weak complementarity, implies that publicly accessible outdoor spaces have not become scarcer, consistent with sustainable management nationally.

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1 Introduction

The outdoors is an important part of American identity and culture, and the value of outdoor recreation is regularly invoked to justify public spending on parks, public lands, and conservation (Banzhaf, 2023). Yet the United States does not keep systematic statistics on Americans' use of the outdoors or the value of that use. Almost no estimates of recreation's non-market value, especially those that are unrestricted by pre-determined sets of activities, are repeated at the national scale through time. This leaves decision-makers unable to track whether the value of outdoor recreation is rising or falling, a necessary input for assessing whether the underlying natural capital is being sustainably managed (Office of Science and Technology Policy et al., 2023; Fenichel, 2024; United Nations Statistical Department, 2025). Inventorying the land and water where recreation occurs is difficult and is missing from current land-use accounts (Wentland et al., 2020). This is partially because trips occur at a diffuse set of locations. These locations, which may enable the majority of Americans' outdoor time, are a mix of natural and built capital, owned and managed by federal, state, and local jurisdictions. Despite these measurement challenges, having regular national statistics on outdoor recreation and its value to Americans is important for assessing the adequacy of investment in accessible outdoor spaces, the sustainable management of those lands and waters, and the public value of that space (Arrow et al., 2004; Dasgupta, 2021).

Here, we use the American Time Use Survey (ATUS) to observe a representative sample of how U.S. residents spend time outside, estimate the willingness to pay (WTP) for those recreation trips, and show how the WTP has changed over time. We then use our WTP estimates to assess whether the accessible outdoor recreation spaces were managed sustainably between 2003 and 2023. The ATUS enables us to construct this time series because the ATUS is a nationally representative time-diary survey in which respondents report every activity over 24 hours, along with the location and duration of each activity. It therefore lets us observe outdoor recreation trips, and the activities that make them up, without deciding in advance which activities or sites count. The ATUS diaries also let us observe each trip's travel time directly. Finally, the ATUS has been run nearly daily since 2003, with a brief pause during the COVID pandemic in 2020.

We find that outdoor recreation trips are overwhelmingly local and built around everyday activities. The implication is that the vast majority of Americans' time spent outdoors is likely within city boundaries, occurring at a diffuse, hard-to-observe set of sites such as public parks, sidewalks, corner parks, schoolyards, and locations that mix natural and built infrastructure. After appealing to weak complementarity, our methods allow us to value this diffuse set of accessible outdoor spaces and to evaluate whether, in aggregate, they have been

managed sustainably, without direct observation of the individual locations.

Economists typically study outdoor recreation by first choosing a site, set of sites, or activity, then defining a study population and measuring how it uses those sites or engages in that activity. Americans seldom pay or "sign in" to use public outdoor spaces, and there is a long tradition of zero-fee access. Therefore, survey data have been critical for measuring use of the outdoors and then estimating value (Lupi et al., 2020). There is an extensive literature on recreation behavior and demand (Oregon State University, College of Forestry, 2025). The literature has informed benefit-cost analyses regarding the conservation or remediation of natural spaces, specifically about water quality (Keiser, 2019; Griffiths et al., 2012; Egan et al., 2009; Bockstael et al., 1987), oil spills (English et al., 2018; Hausman et al., 1995), infectious disease exposure (Day, 2020; Berry et al., 2018), agricultural practices (Hansen, 2007; Sullivan et al., 2004; Hansen et al., 1999), and species conservation (Gürlük and Rehber, 2008; Loomis et al., 2000; Dundas et al., 2018). But these studies are typically site-specific, point-in-time, or both (Lupi et al., 2020), and most are built around iconic pursuits (*e.g.*, fishing or hiking), and thus miss the ordinary trips that we find dominate outdoor recreation. Additionally, these studies frequently rely on on-site intercept surveys or mail and internet surveys sampled from fishing and hunting license holders. Scaling that kind of survey nationally, and repeating it over time, would be prohibitively costly and would still miss trips outside the predetermined sites and activities subject to the survey.

Complementing academic research is the U.S. Bureau of Economic Analysis and Department of the Interior's Outdoor Recreation Satellite Account (ORSA), which tracks spending on complementary market goods for recreation over time (Highfill et al., 2018). ORSA spending, however, is a mix of gasoline and large-ticket items (*e.g.,* recreational vehicles). Therefore, it may follow general spending patterns rather than changes in access to or the quality of places that enable recreation. ORSA also excludes travel expenditures for trips less than 50 miles from home, and our findings suggest the vast majority of outdoor recreation in America falls below this threshold. Yet most local recreators do drive. Seventy-nine percent traveled by car on the day of their trip, so the gasoline spending associated with this recreation is largely invisible in the ORSA.

A separate literature has developed theories for adjusting WTP estimates for non-market goods over time, but has lacked repeated WTP estimates to test their validity. Drupp et al. (2024, 2025) show that an ecosystem service's per-unit value rises as the service becomes scarce relative to produced goods, at a rate governed by the income elasticity of WTP for that service. Testing that framework requires repeated, comparable, national estimates of the same non-market price over a long horizon.

This article makes three contributions. First, we provide a national portrait of how

Americans recreate outdoors, without selecting on activity or site ex ante. Because the ATUS provides a representative sample of all activities Americans engage in, it is feasible to extract an unbiased set of outdoor activities. We use the ATUS to find that most outdoor recreation consists of local trips for everyday activities, a type of outdoor trip largely absent from the recreation valuation literature. For about 80 percent of all outdoor recreation trips, respondents state that the primary purpose is to play with kids or pets, walk, or socialize, and 95 percent require an hour or less of round-trip travel time, which is our definition of local. These frequent activities could all take place at indoor locations close to home. That people nonetheless choose to do them outdoors reveals a preference for accessible outdoor space.

Second, we produce a repeated, national-scale valuation of these local outdoor recreation trips. Because these trips are weakly complementary to the spaces that host them, we recover the value of access to those spaces (Freeman et al., 2014; Phaneuf and Requate, 2017b). Using a discrete choice travel cost model estimated separately over seven three-year periods, we find that the point estimate for WTP per local trip remained stable and was consistently between \$5 and \$6 (real 2023 USD). In contrast, the national value rose 63 percent, from \$57.0 billion to \$93.2 billion over our study period. Growth in national value is driven by an increase in per-person trips and population growth, rather than by large changes in real WTP. The average annual number of local recreation trips taken nationally rose from 10.0 billion in 2003–2005 to 17.4 billion in 2021–2023. This increase is not only population growth, as the annual number of trips per U.S. resident rose from 44 to 65 over the same period. For context, the U.S. Forest Service reports that the National Forest System receives on the order of 148 million recreation visits annually (Rosenberger et al., 2017).

Third, our series of WTP estimates for a local recreation trip lets us empirically test the Drupp et al. (2024, 2025) relative price change (RPC) framework. We first use the framework and the published income elasticities (Drupp et al., 2025) to project our 2003–2005 per-trip WTP forward, assuming no change in the enabling assets. Under standard elasticities, the projected path is consistent with our estimates through 2018–2020 but overshoots our 2021–2023 estimate, because per-trip WTP remained stable while consumption grew. Because the per-trip value is the price of the service flow from accessible outdoor space, we then use the framework and our estimated WTP path to calculate the implied growth of those spaces and find that it is positive. Together with the stable per-trip WTP, rising trip counts, and non-increasing travel times, the framework provides converging evidence that the dispersed natural and built infrastructure supporting local outdoor recreation has been sustainably maintained at the national level over the past two decades.

2 The Character and Distribution of Outdoor Recreation in the United States

We use the ATUS, a nationally representative time-diary survey conducted by the U.S. Bureau of Labor Statistics and widely used in economics, to characterize local outdoor recreation in the United States (Flood et al., 2024). Respondents report the activities they participated in within 24 hours, a categorized "where" location, and the duration of each activity. Each response can be linked to the Current Population Survey, which provides participants' demographic information. The ATUS is a repeated cross-sectional survey, and individuals are only interviewed once.

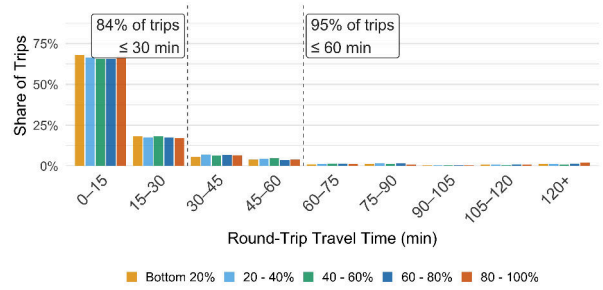
We identify outdoor recreation as leisure activities (outside of work) that take place "Outdoors – not at home," yielding 97 six-digit activity codes that we aggregate into nine specific categories plus an "Other" category. Because respondents report only that an activity took place outdoors and away from home, we do not directly observe the site visited. However, from the activities that took place outside, we can infer the types of places they visit range from natural spaces such as woods, lakes, and beaches to built spaces such as sidewalks, playgrounds, and ball-fields, as well as locations that are a combination of nature and built recreational infrastructure. This mix is not unique to our setting. Recreation demand studies have long valued sites whose appeal reflects a combination of natural attributes and built amenities such as walking paths, boat launches, parking, and visitor facilities (Heagney et al., 2018; Schirpke et al., 2018; Czajkowski et al., 2024).

The aggregated categories, the full activity list, and the composition of trips by income, race, age, and metro status are reported respectively in Appendix A (Tables A1 and A2, Figure A1). We assign each trip to the category of its longest activity. The most common outdoor trips are Playing with Pets (36.5 percent of trips) and Walking for Exercise or Recreation (22.9 percent), followed by Playing with Children (10.7 percent) and Socializing and Relaxing (9.5 percent). Together, these four everyday activities account for about 80 percent of all outdoor recreation trips. Stereotypical outdoor recreation pursuits such as Fishing and Hunting (2.1 percent), Hiking, and Skiing account for far fewer trips. The ranking changes when activities are instead ranked by annual hours per U.S. resident. Walking (11.1 hours) and Playing with Pets (10.8 hours) remain the top two, but Socializing and Relaxing (7.6 hours) moves ahead of Playing with Children, and Fishing and Hunting (4.9 hours) and Sports (4.6 hours) rise to match or exceed Playing with Children (4.6 hours). Socializing, hunting and fishing, and sports trips last longer on site than a trip to take a child outside to play, and hunting and fishing trips in particular are infrequent but long.

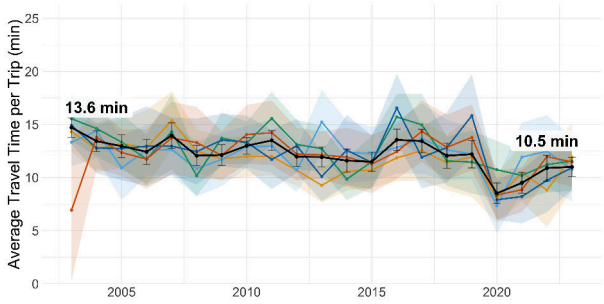
Trip composition varies substantially across demographic groups (Appendix Figure A1).

Figure 1: Travel Time, Trip Frequency, and On-Site Time of Local Outdoor Recreation by Income Quintile, 2003–2023

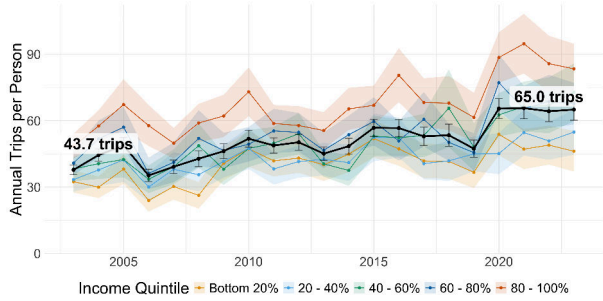
(A) Distribution of Round-Trip Travel Time



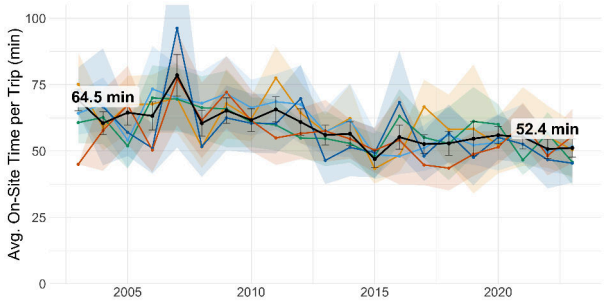
(B) Average Round-Trip Travel Time



(C) Annual Local Recreation Trips per Person



(D) Average On-Site Time per Trip



Most outdoor recreation happens close to home. Since 2003, local trips have become more frequent and shorter, with less travel per trip, in every income quintile. Panel (A): share of each quintile's outdoor recreation trips (local and non-local) in each round-trip travel-time bin; bins include their upper bound, and the dashed lines mark the 30- and 60-minute thresholds. Panels (B)–(D) cover local trips only (round-trip travel time $\leq$ 60 minutes) and plot annual weighted averages by income quintile, with the black line showing the national average: (B) round-trip travel time per trip; (C) trips per U.S. resident (denominator: full population, including non-recreators); (D) on-site time per trip (dwell time). Colors follow the legend in Panels (A) and (C). All estimates use ATUS sample weights. Shaded bands and error bars in (B)–(D) are 95% confidence intervals from ATUS replicate-weight standard errors. Labels on the black line report national averages for 2003–2005 and 2021–2023.

Playing with Pets and Running increase as a share of trips with income. Playing with Children, Socializing/Relaxing, and Walking decline with income, possibly reflecting yard access: higher-income households may conduct these activities at home, while lower-income households without yard access use parks and public spaces. Playing with Pets is the top activity for White people (non-Hispanic; 40.5% of trips), while Walking (28.1%) narrowly leads Playing with Pets (26.6%) for Black people, who also devote a larger share of trips to Socializing and Relaxing (13.5%) than White people (8.6%). Hispanic people devote a much larger share of outdoor trips to Playing with Children (20.4% of trips) than White people (8.4%). Trips to socialize and relax, run, or play sports decrease in frequency with age, whereas walking increases, as does playing with pets through middle age. Fishing and

Hunting makes up a far larger share of trips for rural Americans than for those living in an urban core or suburb.

We calculate the travel time for outdoor recreation trips using individuals' daily diaries. A trip is an outing: consecutive outdoor activities, including those separated only by travel between stops, are merged into a single trip. We sum the durations of the nearest travel activity before the trip's first activity and the nearest travel activity after its last activity (each within two diary entries), plus any travel between stops, to obtain the round-trip travel time. Travel time can equal zero for trips that begin and end at an individual's home, such as a walk from the front door. We exclude outdoor activities recorded as taking place inside a vehicle (*e.g.,* eating while driving). We define a trip as "local" if its round-trip travel time is one hour or less. This threshold matches the time Americans spend traveling daily for work, as the average metropolitan round-trip commute was 52 minutes at the midpoint of our study period (U.S. Census Bureau, 2017). Eighty-four percent of outdoor recreation trips require thirty minutes or less of round-trip travel time, and 95 percent require an hour or less. There is no major variation in the distribution of travel time when examined by separate income quintiles (Figure 1, Panel A) or racial groups (Appendix Figure A2). Travel time does, however, vary by the type of activity: trips for Fishing and Hunting take the longest on average, while Walking and Playing with Pets require the least travel (average per-trip travel time by activity is reported for grouped activities in Appendix Table A1 and for disaggregated activities in Appendix Table A2). The activities that dominate trip-taking are also the cheapest to access. It is therefore their frequency that drives their outsized role in the aggregate value of local recreation.

The average round-trip travel time of a local recreation trip decreased from 13.6 minutes in 2003–2005 to 10.5 minutes in 2021–2023 (Figure 1, Panel B). The decline is not a steady trend. Annual averages fluctuated between 11.5 and 14.0 minutes from 2004 to 2019, fell to 8.5 minutes in 2020, and had partly recovered to 11.0 minutes by 2023. An overall decline across all recreators drives this decrease. It is not attributable to significantly different travel time per trip for any specific income quintile or racial group (Appendix Figure A2). It is also not driven by urbanization. While the U.S. population has become more urban (rural share: $19.9\% \pm 0.42$ SE in 2003–2005 vs. $12.8\% \pm 0.36$ SE in 2021–2023), rural and urban recreators have statistically indistinguishable average travel times (urban/suburban: 12.0 ± 0.12 SE min; rural: 12.1 ± 0.35 SE min).

The average number of local trips per U.S. resident increased from 43.7 per year in 2003–2005 to 65.0 in 2021–2023, a 49 percent increase (Figure 1, Panel C). This increase occurred across all income quintiles, though it is most pronounced among the highest earners. The lowest income quintile saw a significant increase in the average number of annual trips

taken between 2008 and 2010, around the time of the U.S. financial crisis. The highest income quintile experienced a similar but larger increase between 2019 and 2020, during the COVID-19 pandemic. In both cases, the increase in annual trips persists over time, indicating lasting shifts in behavior.

While trips have become more frequent, they have also become shorter. The average on-site time spent at the recreation activity itself fell from 64.5 minutes per trip in 2003–2005 to 52.4 minutes in 2021–2023 (Figure 1, Panel D). As with travel time, this decline is broad-based across income quintiles and racial groups (Appendix Figure A2) rather than concentrated in any one group. Industry reports document that the number of outdoor participants has grown, but the typical participant now takes fewer trips (Outdoor Industry Association, 2022). Our diary-based measure tells a different story: Americans are taking more frequent trips to closer sites, but they spend less time at their destinations on average. The increase in trips taken is the primary driver of the rising national value of local outdoor recreation that we document.

3 Methods

We estimate the willingness to pay (WTP) for a local outdoor recreation trip with a discrete-choice travel cost model. We then use the resulting 20-year series of per-trip WTP estimates to test whether using the RPC framework to adjust the WTP for local recreation at the beginning of the time series yields a calculated WTP series similar to the one we empirically estimate. In this section, we state the assumption that connects trip-taking to the value of the underlying public assets, then describe how we construct our recreation choice data, present the travel cost model and our identification strategy, and explain how we calculate the WTP for a trip, how we aggregate that value to the personal and national level, and how we use the WTP estimates to test the RPC framework.

Three threats to identifying the WTP for a local recreation trip guide our estimation approach. First, we observe travel time for respondents who happen to take a recreation trip on their diary day, but we cannot directly observe the counterfactual travel time of individuals who do not take a trip on their diary day. We refer to these individuals as non-recreators, though the label describes only their diary day. We recover the expected counterfactual travel time using nearest-neighbor propensity-score matching to recreators facing a comparable set of recreation opportunities. Carefully constructing this counterfactual travel time allows us to identify how travel cost influences the probability of taking a trip, which is a core parameter for later calculating the WTP for a trip. Second, the travel cost for a trip may be partly monetary (*e.g.,* gasoline and admission fees), and we do not

observe individuals' monetary expenditures, raising concerns about omitted variable bias. Because local recreation is typically free at the point of use, the monetary expenditure of concern is primarily the cost of driving; 79 percent of local recreators traveled by car on their diary day. In Section 3.3, we outline how we control for the regional real price of gasoline, interacted with car use, so that variation in the monetary cost of driving does not bias the travel-cost coefficient. Third, people may sort such that low observed travel costs partly reflect location choice rather than the behavioral response to cost; for example, "recreation-lovers" may choose to live near recreation sites. We consider how sorting may be a concern in this setting, how our controls absorb its observable component, and when residual sorting will bias our WTP estimates downward, making them lower bounds.

3.1 Weak Complementarity: Value of Public Asset

Environmental economists have appealed to weak complementarity for over 50 years to value environmental assets using revealed-preference techniques such as travel cost models (Mäler, 1974). In the canonical travel cost study, the value of a specific environmental amenity (*e.g.,* a lake, a forest, or a stretch of coastline) is estimated using observed trips to the site. The value recovered is that of the site as a bundle of its natural amenities and the built infrastructure that makes it accessible and improves its quality (Heagney et al., 2018; Czajkowski et al., 2024). Even models of iconic natural sites such as Yellowstone National Park treat visitor welfare as depending on park infrastructure in addition to the natural ecosystem (Settle et al., 2002). Variation in travel cost comes from variation in where individuals live, and the model estimates how trip-taking declines as the cost of access rises. Weak complementarity links this trip-taking behavior to the site's value and assumes that the amenity contributes to utility only through the trips it enables. A person who never visits is indifferent to changes in its quality or existence. Under this assumption, the consumer surplus derived from trip-taking behavior measures the use value of the environmental asset (Phaneuf and Requate, 2017a).

We assume that the ordinary, frequent outdoor trips documented in Section 2 can be used to estimate the value of the diffuse collection of local accessible outdoor spaces that host them. As in the site-based literature, weak complementarity enables estimation of the value of these public outdoor spaces from trip behavior. The consumption good is any local outdoor recreation trip observed in the ATUS without a predetermined list of activities or sites. This differs from the canonical case only in that the asset we value is diffuse, rather than a specific, identified location. Weak complementarity holds in our setting because all observed trips in our dataset occur "Outdoors – not at home," so they are only possible in

an accessible outdoor space. Because the consumption good is a trip to any local outdoor space, our model allows substitution among outdoor sites. A recreator who cannot use one park can travel to a nearby greenway or neighboring park instead. Weak complementarity therefore applies to the complete set of local sites, and our WTP measures the value of access to that collection rather than to any individual location. This also means that we assume a person who does not take such a trip derives no utility from the recreation channel in those spaces and is thus indifferent to these sites' condition for recreation.

Our estimates for these public outdoor spaces are conservative. As in any travel cost model, we recover only the recreational use value, which is the flow of value from trip-taking. Many outdoor spaces also provide services that do not require a trip, including improved air quality, stormwater drainage, noise reduction, and heat mitigation (Heris et al., 2021). The value of these indirect-use services is additive to recreational use value but is not captured by any travel cost model, so our estimates, like those from site-based studies, are lower bounds on the total value of the local outdoor spaces that enable recreation.

3.2 Constructing Recreation Choice Data

Estimating a travel cost model requires data on whether a trip was taken and the travel cost of that trip for recreators and non-recreators alike. The ATUS provides both for respondents who take a local outdoor recreation trip on their diary day, because we observe the trip itself and its realized round-trip travel time. In the ATUS, each respondent's diary is a minute-level, sequential account of the entire day. Travel is recorded as its own activity within these daily-diary logs, and we link it to the trip it enables. We use the sequential aspect of the ATUS's daily diaries to define each trip as an outing: consecutive outdoor recreation activities, including those separated only by travel between stops, form a single trip. We attribute to each trip the nearest travel activity within two diary entries before its first activity and the nearest travel activity within two diary entries after its last activity, plus any travel between its stops. This yields the realized round-trip travel time of that specific recreation trip without assuming distances, travel speeds, or that the trip began at home.

For respondents who do not take a trip, we observe the choice not to recreate but not the travel time they would have faced. We impute this travel time using nearest-neighbor propensity-score matching, pairing each non-recreator with recreators from the same U.S. state and three-year period who had similar time constraints on their diary day. Matching within state and period holds the state-level supply of recreation opportunities fixed. The propensity score matches non-recreators with recreators who have similar time availability

so that matched recreators face a similar set of recreation opportunities. The propensity score uses time-use and calendar variables together with non-metropolitan residence, age, and sex. All these variables either constrain the decision to take a trip or are determined before it. We select the propensity-score equation and a caliper of 0.25 to balance covariates between non-recreators and their matched recreators, and use $n = 20$ matched neighbors, a value in the range over which WTP per trip is stable. We also use the matched neighbors to impute the type of outdoor activity a non-recreator would have done had she taken a trip, measured as the share of her matched neighbors who participated in each activity category. A placebo test that imputes travel times for recreators, whose travel times we observe, shows that matching recovers average travel time (18.4 minutes imputed versus 18.1 minutes observed; Appendix C4). Because each imputed value is an average over matched neighbors, however, imputed travel times cluster around 15 minutes, whereas 41 percent of observed trips involve no travel time; this difference in the shape of the distribution is an artifact of imputing, rather than observing, travel time for non-recreators. Appendix C details the propensity-score specifications and balance diagnostics, and shows how WTP per trip responds to the number of matched neighbors and the caliper.

3.3 Discrete Choice Travel Cost Model

We follow McFadden's binary random utility model (McFadden, 1974a,b). Each respondent i chooses between taking an outdoor recreation trip ($j = 1$) and a "no-deviation" day that holds all other activities fixed ($j = 0$). The reason we consider a no-deviation day, rather than a no-trip day, is that the way we identify an individual's travel time allows travel from a location other than home. For example, if an individual were to take a multi-purpose trip in which she traveled from the store to a local park and then from the park to home, we would include the travel time from the store to the park and from the park to home as her round-trip travel time.

The utility that respondent i obtains from choice j is

$$U_i^j = V_i^j + \varepsilon_i^j,$$

where V_i^j is the deterministic component of utility, a function of observables, and ε_i^j is an error term that we assume follows a Type I Extreme Value distribution. This distributional assumption yields logistic choice probabilities.

Because local outdoor recreation is typically free at the point of use, the price of a trip is dominated by the cost of getting to the site, which has two components: travel time and the monetary cost of the trip. Travel time is the primary component, and we observe it directly

in the time diaries. We monetize travel time using the opportunity cost of time, which we set to one-third of the wage following convention (Lupi et al., 2020). Because the ATUS does not report a wage for every respondent, we use the average hourly wage implied by family income, computed as annual family income divided by 2,080 hours (a replacement-wage alternative is in Appendix B3).

The monetary component, primarily gasoline for a local trip, is not observed at the individual level. Because car travel is common among recreators and the monetary cost of driving is likely correlated with travel time, this missing expenditure is an omitted variable that could bias the estimated response to travel cost. We cannot observe individual expenditures, so we instead control directly for the regional real price of gasoline, which varies across regions and months.

We condition on seven three-year periods between 2003 and 2023, the finest resolution at which the ATUS supports reliable national inference of this behavior. Pooling three adjacent years per period provides enough recreators to estimate the model with the full set of controls, while estimating each period separately lets the unobserved supply of recreational opportunities evolve over time. Normalizing $V^0 = 0$, we model the indirect utility of a trip as

$$\begin{aligned} V^1_{i,r,m} \mid t = {} & \beta_0 + \beta_1\, \pi_i + \boldsymbol{\beta_2}\, \mathbf{State}_i + \boldsymbol{\beta_3}\, \boldsymbol{D}_i + \beta_4\, y_i + \boldsymbol{\beta_5}\, \boldsymbol{X}_i \\ & + \beta_6\, \mathrm{Car}_i + \beta_7\, L_i^{\mathrm{eat}} + \beta_8\, G_{r,m} + \beta_9\, G_{r,m} \times \mathrm{Car}_i, \end{aligned} \tag{1}$$

where π_i is the travel cost of the trip (travel time monetized at one-third of the wage), $\boldsymbol{D}_i$ is a vector of outdoor activity types (nine categories with walking omitted; imputed for non-recreators as described in Section 3.2), y_i is family income in 2023 USD, $\boldsymbol{X}_i$ is a vector of demographic characteristics (race/ethnicity, age, rural residence, and sex), Car_i indicates whether the respondent traveled by car on the diary day, and L_i^{eat} is the average travel time to restaurants in the respondent's county (when available, otherwise state[1]) and year. L_i^{eat} is a proxy for the local availability of an indoor leisure substitute. State fixed effects control for cross-state differences in park supply, climate, and the culture of outdoor recreation (full variable definitions in Appendix B1).

The final two terms address the unobserved monetary cost of the trip. $G_{r,m}$ is the monthly, m, real retail price of gasoline in the Petroleum Administration for Defense District (PADD, r) containing respondent i's state of residence in the month of her diary day (U.S. Energy Information Administration, 2025). Its interaction with Car_i allows gasoline prices to directly affect those who likely drive. Because the gasoline terms are not interacted with travel cost in our preferred travel cost model, the travel-cost coefficient β_1 is identified from residual

[1]County is not available for respondents who live in locations with fewer than 100,000 people, in which case we observe only their state.

variation in travel time after controlling for variation in the monetary cost of driving. As a further robustness check on the monetary-cost channel, Appendix B2 reports an additional specification that interacts the gasoline price with travel cost itself, which allows the marginal disutility of travel cost to vary with the price of gasoline.

3.4 Identification

We identify the marginal utility of travel cost, β_1, from variation in travel cost, π_i, across individuals. Three features justify using this variation for identifying a causal effect. First, we observe travel time directly from respondents' time diaries, rather than constructing it from assumed distances or speeds, avoiding measurement-error concerns about travel time (Bradt, 2025). Second, we address the omitted-variable concern from unobserved monetary trip expenditures by controlling for the regional real price of gasoline and its interaction with car use, as described in Section 3.3, so that the monetary cost of driving does not lead to omitted variable bias when estimating β_1. Third, we impute the travel costs of non-recreators by matching them to recreators in the same state and period. Identification, therefore, rests on comparisons among observationally similar people who face the same local supply of recreational opportunities.

The main remaining threat is residential sorting. A large literature in environmental economics shows that households incorporate local environmental amenities into their location choices (Bayer et al., 2009; Klaiber and Phaneuf, 2010; Kuminoff et al., 2013), and recent work demonstrates that the resulting correlation between unobserved recreation tastes and travel costs can bias recreation demand estimates in either direction (Bradt, 2025). When recreation-lovers seek recreation access, low travel costs predict trip-taking because people choose to live near parks. The estimated price response overstates price sensitivity and biases WTP per trip downward. Sorting away from recreation sites reverses the bias, leading to overestimates of WTP, as in Bradt (2025).

Three features of our setting mitigate the concern that sorting biases our results. First, sorting operates through the choice of where to live, whereas our outcome is whether an individual took a local recreational trip on the day she was surveyed by the ATUS. Given where a respondent lives, the travel cost she faces on her diary day reflects the idiosyncratic realization of which site she visited and how far away it was, rather than a systematic measure of her taste for recreation. This is the canonical assumption that justifies treating the residual variation in travel cost π_i as exogenous in recreation demand models. Second, the amenity we value is diffuse. Local parks, sidewalks, and schoolyards are found in the majority of neighborhoods, unlike destination amenities (*e.g.,* a coastline, a national park, a named lake

with campsites) for which sorting models are built. Sorting pressure is weak when the asset is ubiquitous, and estimated preferences for local parks are small and relatively homogeneous across household types, in contrast to regional parks and other destination amenities whose values are large and heterogeneous in sign (Klaiber and Phaneuf, 2010). Third, the controls in Equation 1 control for the observable channels through which sorting operates. State fixed effects capture cross-state differences in park supply, climate, and recreation culture. Income, y_i, and $\boldsymbol{X}_i$ control for sorting on income, race, life stage, and sex. Rural and car indicators capture within-state differences in access and the available choice set, and the gasoline terms hold the monetary cost of driving fixed. The activity controls $\boldsymbol{D}_i$ ensure we identify the travel-cost response within, not across, activity types.

We do not claim to eliminate sorting on unobserved tastes among observationally identical respondents, and the sign of any residual bias is unknowable ex ante because it depends on the direction of sorting on unobservables. However, if sorting on unobservables follows the conventional direction emphasized in this literature, with strong recreation tastes causing "recreation-lovers" to sort near outdoor access, then residual sorting leads us to overstate price sensitivity. In that case, our WTP estimates are lower bounds on the value of access to local outdoor space.

3.5 Willingness to Pay per Trip

Under the Type I Extreme Value assumption, the probability of taking a trip is logistic,

$$\Pr(j=1) = \frac{\exp(V^1)}{\exp(V^0)+\exp(V^1)} = \frac{\exp(V^1)}{1+\exp(V^1)},$$

and we estimate the parameters of Equation 1 by maximum likelihood. An individual's value of a daily trip is the difference between the expected maximum utility when a trip is available and the utility when it is not. Under the Type I Extreme Value assumption, $\mathbb{E}[\max_j(V^j+\varepsilon^j)] = \ln\sum_j e^{V^j}$ up to an additive constant that differences out (Small and Rosen, 1981), so this difference reduces to the log-sum of indirect utilities. Converting to dollars using $-1/\beta_1$ and applying $V^0 = 0$ gives the expected value of the option to take a trip on a given day,

$$\mathrm{CS}_{i,t} = -\frac{1}{\beta_1}\ln\left(e^{V_i^1}+1\right),$$

where t is the three-year period that individual i was surveyed within, and V_i^1 is the indirect utility of a trip from Equation 1, which equals the log-odds of taking a trip, $\ln\big[\Pr_i(j=1)/\big(1-\Pr_i(j=1)\big)\big]$, because $V^0 = 0$. Because this daily value accrues to everyone with access, whether or not they take a trip that day, we express it per trip by dividing by the

probability of taking a trip. The average WTP per trip in period t is

$$\overline{\text{WTP}}_t = \frac{\sum_i w_i \widehat{\text{CS}}_{i,t}}{\sum_i w_i \widehat{\Pr}_i(j=1)}, \tag{2}$$

where w_i is the ATUS sampling weight and each respondent's diary day is counted once. We compute standard errors for $\overline{\text{WTP}}_t$ by the delta method applied to this ratio, using the coefficient covariance matrix clustered by respondent.

3.6 Aggregating to Per-Person and National Values

We calculate per-person and national annual values by scaling the average per-trip WTP by the relevant trip count, and calculate standard errors via the product rule. The per-person annual value of access to local outdoor recreation is

$$\Upsilon_t^{\text{personal}} = \overline{\text{WTP}}_t \cdot \mathbb{E}[x_{i,t}], \tag{3}$$

where $\mathbb{E}[x_{i,t}]$ is the expected number of local outdoor recreation trips per person per year in period t, computed following the Bureau of Labor Statistics' (BLS) methodology for ATUS-based annual estimates (U.S. Department of Labor and Bureau of Labor Statistics, 2022). Incorporating the BLS's methodology ensures we correctly calculate average behavior for U.S. residents, and is important for incorporating the sampling uncertainty from the ATUS sample in our per-person and national estimates. The national annual value is

$$\Upsilon_t^{\text{national}} = \overline{\text{WTP}}_t \cdot X_t, \tag{4}$$

where X_t is the total number of local outdoor recreation trips taken in period t, computed by summing ATUS sampling weights over local recreation trips. We estimate standard errors for X_t and $\mathbb{E}[x_{i,t}]$ using the ATUS replicate weights. This allows us to propagate the WTP and trip-count uncertainty via the product rule when assuming sampling variation in trip counts is independent from estimation uncertainty in $\overline{\text{WTP}}_t$. Aggregating individual-level WTP to a national value also assumes local outdoor recreation is a homogeneous good, after conditioning on our controls (Addicott and Fenichel, 2019).

3.7 Testing the Relative Price Change Framework

Our 20-year repeated measure of per-trip WTP in real 2023 USD lets us empirically test the relative price change (RPC) framework of Drupp et al. (2024), which predicts that the real

price of an ecosystem service evolves as

$$\text{RPC}_t = \xi\,(g_{C,t} - g_{E,t}), \tag{5}$$

where ξ is the income elasticity of WTP, g_C is the growth rate of real consumption per capita, and g_E is the growth rate of the underlying ecosystem service flow. Per-trip WTP is the relevant unit price because using aggregate values would reflect growth in the quantity of trips rather than the non-rival and non-excludable asset's scarcity.

We use Equation 5 in two ways. First, we use our estimate of WTP in the 2003–2005 period and project that WTP forward 20 years. This allows us to test whether the RPC framework's projected path aligns with our estimated $\overline{\text{WTP}}_t$ series. Second, we solve Equation 5 for g_E to back out the implied asset growth from the observed changes in WTP, which we relate to relative price changes. The detailed methodology is presented in Appendix D1. We do not estimate ξ. We instead use four benchmarks as a sensitivity range ($\xi \in \{0.60, 0.74, 0.86, 1.91\}$ from those collected by Drupp et al. (2025)). We use real personal consumption expenditures per capita for g_C. Because real personal consumption expenditures are not as commonly available, we also use real GDP per capita as a robustness check (details and results in Appendix D3). Full derivations, the ξ benchmarks, and data sources are in Appendix D.

Weak complementarity allows us to use our per-trip WTP series as the input to the RPC framework. The framework considers the price of an ecosystem service flow and how that price is expected to move as the underlying asset becomes more or less scarce relative to produced goods. If the asset gets scarcer, the flow's price rises, and vice versa. In the accounting language of Boyd and Banzhaf (2007), an ecosystem service is the final flow directly enjoyed by people, and that flow is generated by its enabling natural asset. In our setting, this final service is local outdoor recreation, and the asset is the stock of accessible outdoor space that hosts it. Our per-trip WTP is the price of the service these spaces enable. So if accessible outdoor space had become scarcer over our study period, we would observe an increase in WTP per trip. If it had become more abundant, WTP per trip would have fallen. Our 20-year series of per-trip WTP therefore lets us test whether the price of this service has evolved as the framework predicts.

4 The Value of Outdoor Recreation in the United States

Estimating the travel cost model (Equation 1) on all years pooled (2003–2023) yields an average WTP for a local outdoor recreation trip of \$5.6 (SE: \$0.2) in real 2023 USD

(Appendix Table B2). We interpret this as a lower bound on the value of these local outdoor spaces. Our estimates capture only their recreational use value, not the additional ecosystem services these spaces provide beyond recreation (see Section 3.1). In addition, if individuals with stronger tastes for recreation choose to live near such spaces, sorting biases our estimates downward (see Section 3.4).

The estimate is stable in the number of recreators matched to each non-recreator to impute travel cost. At our preferred caliper (0.25), it varies by less than one standard error for 15 to 25 matched recreators (Appendix C3). It is also robust to allowing the monetary and time costs of a trip to interact. Our specification that interacts the gasoline price with travel cost leaves the average travel-cost response and per-trip WTP statistically unchanged (Appendix B2, Model 12). Therefore, we report welfare estimates from the simpler model without a travel cost interaction term, as described in Equation 1.

Re-estimating the model separately conditional on each three-year period yields per-trip WTP estimates from 2003 to 2023 (Figure 2, Panel A). The per-trip price remained stable, at \$5.7 in 2003–2005 and \$5.3 in 2021–2023, and the confidence intervals of the first and last periods overlap. Period estimates fluctuate within a \$5.2–\$6.0 band around the pooled value of \$5.6, and no three-year window differs significantly from it. That per-trip WTP remained stable over two decades of population, income, and land-use change suggests that local recreation has not become meaningfully scarcer at the margin within the United States, which we test formally in Section 5.

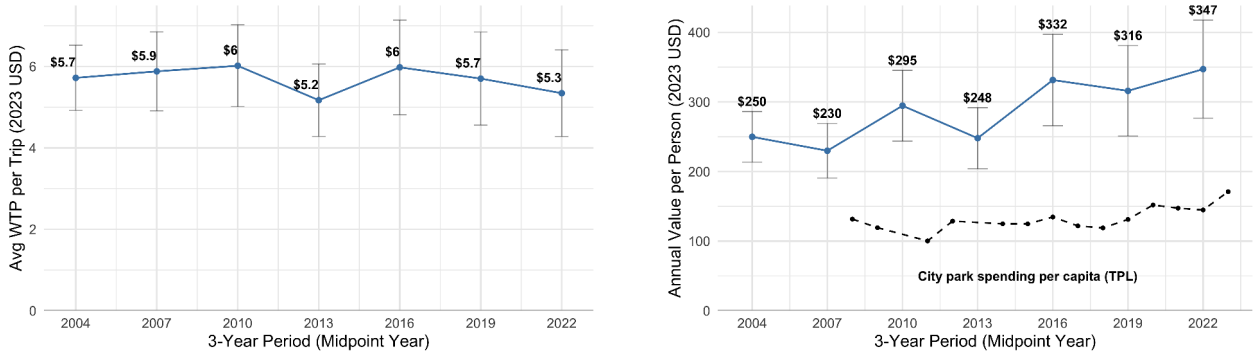
We calculate the annual per-person value of access to local recreation by aggregating per-trip WTP to the annual per-person level (Figure 2, Panel B). This value averaged \$288.1 over the study period and grew by 39 percent, from \$249.8 in 2003–2005 to \$347.1 in 2021–2023. Per-trip WTP remained stable; therefore, the increase in the annual value per person is driven by a rise in the average number of local recreation trips taken annually per person (Figure 1, Panel C). As a benchmark, the annual per-person value of access is roughly twice the annual per capita spending on city parks reported by The Trust for Public Land (2025), which includes monetary spending and the value of volunteer hours. We include this per capita spending as the dashed black line in Figure 2, Panel B.

The annual national value of local outdoor recreation averages \$71.7 billion over the study period and rose from \$57.0 billion in 2003–2005 to \$93.2 billion in 2021–2023, a 63 percent increase (Figure 2, Panel C). The national growth combines the per-person increase, which itself reflects a rising share of U.S. residents taking local recreation trips, with population growth. On an average day in 2003–2007, 9.3% (SE: 0.13%) of U.S. adults took a local outdoor recreation trip; by 2019–2023, that share had risen to 13.2% (SE: 0.20%).

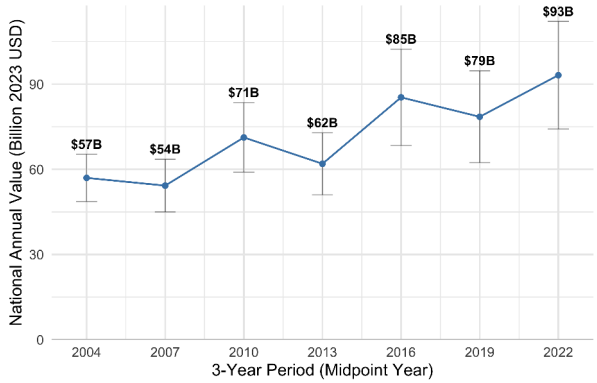
Figure 2 shows WTP estimated using one-third of respondents' average hourly wage,

Figure 2: The Value of Local Outdoor Recreation per Trip, per Person, and Nationally, 2003–2023

(A) WTP per Local Recreation Trip ($\overline{\text{WTP}}_t$) (B) Annual Value of Access per Person ($\Upsilon_t^{\text{personal}}$)



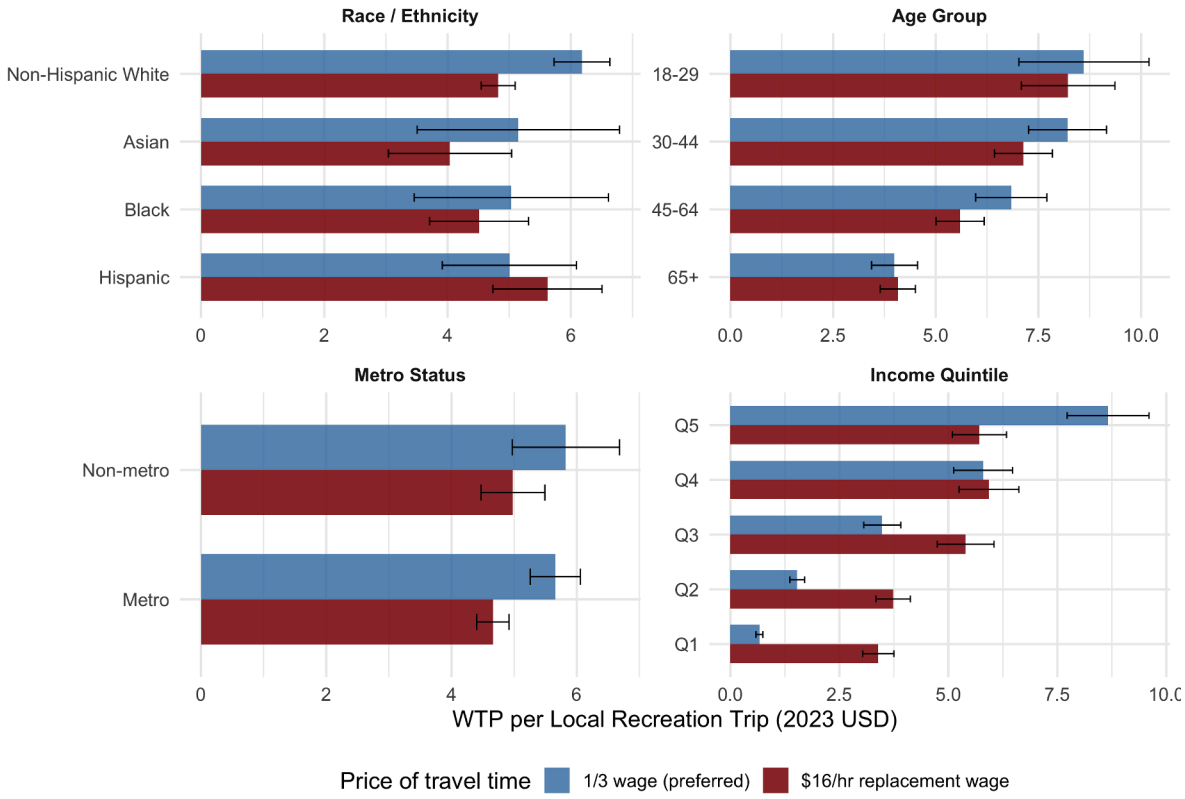
(C) Annual National Value of Access ($\Upsilon_t^{\text{national}}$)



Per-trip WTP remained stable while the annual per-person and national values grew by 39 and 63 percent. Growth came mainly from more trips per person, not a higher value per trip. Panel (A): average WTP per local recreation trip ($\overline{\text{WTP}}_t$, Equation 2). Panel (B): annual per-person value of access ($\Upsilon_t^{\text{personal}}$, Equation 3); the dashed black line is per capita spending on city parks, including the value of volunteer hours (The Trust for Public Land, 2025). Panel (C): annual national value of access ($\Upsilon_t^{\text{national}}$, Equation 4). The model is estimated separately for each three-year period, and each estimate is plotted at the midpoint year of its period (e.g., 2003–2005 at 2004). Travel time is priced at one-third of the respondent's average hourly wage. Error bars are 95% confidence intervals: delta method for (A), and for (B)–(C) a product rule combining the delta-method WTP standard error with the ATUS replicate-weight standard error on trip counts. Real 2023 USD.

the standard travel-cost convention for capturing the opportunity cost of travel time (Lupi et al., 2020). As a sensitivity test of this pricing method, we re-estimate the model using a single replacement wage of $16.08 per hour, the replacement wage used by the Bureau of Economic Analysis to value household production (Bureau of Economic Analysis, 2018), in place of one-third of the respondent's average hourly wage. Average WTP per local recreation trip is somewhat lower when travel time is priced with the replacement wage, but the 95% confidence intervals under the two values of travel time (VOTT) overlap in every

Figure 3: WTP per Local Recreation Trip by Demographic Subgroup under Two Values of Travel Time



How travel time is priced moves average WTP only modestly, but it changes comparisons across groups, especially by income. The top-to-bottom income quintile gap is large when time is valued at one-third of each respondent's average hourly wage and much smaller under a common replacement wage. Each bar re-estimates the full travel cost model (Equation 1) within the subgroup (details in Appendix B4). "1/3 wage" (preferred) prices travel time at one-third of each respondent's average hourly wage; "$16/hr replacement wage" prices it at $16.08 per hour for everyone. Income quintiles come from the family-income distribution in real 2023 USD (Q1 = bottom 20%, Q5 = top 20%). Metro status follows the ATUS metropolitan indicator. Error bars are 95% confidence intervals from delta-method standard errors. Real 2023 USD.

three-year period (Appendix B3).

While the average WTP per trip changes little when the replacement wage is used for the VOTT, the WTP estimates for different demographic groups are sensitive to how travel time is priced (Figure 3). Two patterns stand out. First, White respondents have a meaningfully higher per-trip WTP when using the one-third-wage pricing approach rather than the replacement-wage approach. Second, the highest-income quintile (Q5, the top 20% of the family-income distribution) has by far the largest per-trip WTP under the one-third-wage pricing method. This reflects the mechanical link between wages and the traditional opportunity-cost-of-time calculations. The gap between the top quintile and the

lowest narrows considerably when using a replacement wage. The implication is that average WTP estimates for local recreation change little with how we price travel time in this paper, but distributional comparisons across income and race change substantially. Distributional analyses of the value of outdoor recreation, therefore, require transparency about VOTT assumptions, as the choice of how to measure the opportunity cost of time encodes normative judgments about whose time counts more. Appendix B4 describes the subgroup estimation procedure in detail.

5 The Change in Wealth Provided by Assets Supporting Outdoor Recreation

A benefit-cost analysis conducted today often must rely on a willingness-to-pay estimate from years or even decades earlier because repeated national valuations of an ecosystem service are rarely available. The relative price change framework of Drupp et al. (2024, 2025) is a way to update "stale" values, scaling old estimates forward at a rate determined by the income elasticity of WTP and the changing relative scarcity of the service. However, whether that adjustment is warranted has been difficult to verify because it requires repeatedly valuing the same service over time. Our 20-year WTP series supplies this opportunity, permitting a direct empirical test of the RPC framework for local outdoor recreation, which is a major ecosystem service. Because the outdoor spaces that host local recreation are weakly complementary to the trips taken to them (Section 3.1), the consumer surplus we recover from trip-taking measures the value of access to those spaces. The per-trip WTP is therefore the marginal value of access to the enabling outdoor space, and, within the RPC framework, changes in this unit price over time reflect the land's changing relative scarcity. We find that, under standard income-elasticity benchmarks and no change in the enabling assets, the framework tracks per-trip WTP through 2018–2020 but overshoots it in 2021–2023, because per-trip WTP remained stable while consumption grew. Inverting the framework, we find that the enabling assets for local recreation have become no scarcer over the study period.

Empirical benchmark for g_E. The natural baseline for the growth rate of the ecosystem service flow, g_E, is zero: an analyst who updates a stale WTP estimate using income growth alone implicitly assumes that the enabling assets are unchanged. We therefore use $g_E = 0$ as the preferred value for our test of the RPC framework. Two independent series are consistent with recreation-enabling assets having, if anything, grown. On the supply side, total park

acreage across the 100 largest U.S. cities grew on average 9.8 percent per three-year period from 2010 to 2022 (Appendix Table D3). On the demand side, total national local outdoor recreation trips from ATUS grew on average 10.7 percent per three-year period from 2003 to 2023 (Appendix Table D4). City park acreage expanded in absolute terms, at a similar rate to local recreation participation. Appendix D4 reports sensitivity to alternative g_E values, including aggregate ecosystem service decline at -1.01% and global forest decline at -0.11%, each applied per three-year period (Appendix Figure D3). At the recreation-and-ecotourism income elasticity ($\xi = 0.74$), the RPC framework's projected WTP path stays within our estimates' confidence intervals in every period only if recreation-enabling assets grow by between 1.9% and 8.9% per three-year period (*i.e.,* the framework holds if $g_E \in [1.9, 8.9]$, Appendix Figure D4).

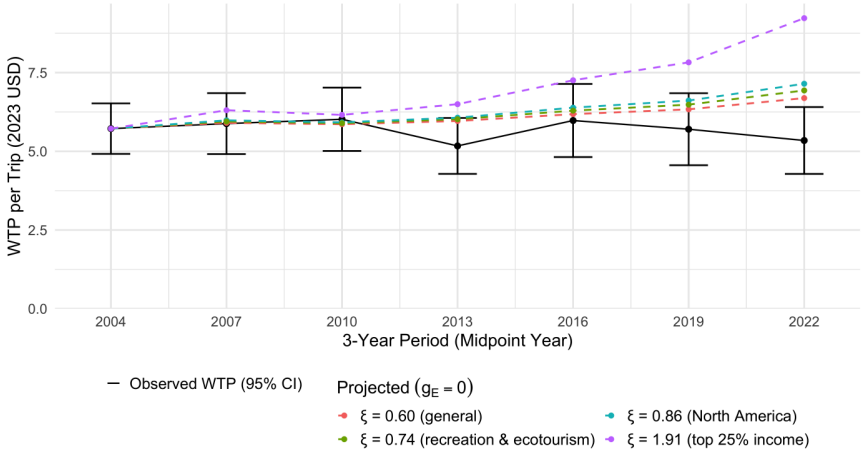
Test: Does WTP grow as the RPC framework predicts? Figure 4 plots the observed per-trip WTP against the projected path of WTP values from the RPC framework under $g_E = 0$, for each of the four ξ benchmarks, using per capita expenditure as g_C (see Appendix Equation 7 for projection path description). The projected paths for $\xi = 0.60$ and $\xi = 0.74$ lie within the 95% confidence intervals of the observed WTP estimates in every period through 2018–2020, and the $\xi = 0.86$ path does as well except in 2012–2014.[2] In 2021–2023, however, all four projections exceed the confidence interval, ranging from \$6.69 ($\xi = 0.60$) to \$9.23 ($\xi = 1.91$), compared with an observed WTP of \$5.34 (95% CI: \$4.28–\$6.41). The highest income elasticity estimate ($\xi = 1.91$), which would make local recreation a strong luxury good, falls outside the intervals from 2012–2014 onward. The projections diverge from our estimates because per-trip WTP remained stable while real consumption per capita grew, so the framework with an unchanged asset stock predicts a rising price that we do not observe. The pattern is the same when consumption growth is measured with real GDP per capita instead; the projections for $\xi \in \{0.60, 0.74, 0.86\}$ lie within the confidence intervals in every period except 2021–2023 (Appendix D3).

Implied asset growth, $\hat{g}_E$. Inverting the RPC framework to solve for g_E delivers a direct estimate of how fast the stock of recreation-enabling assets has grown. The implied average g_E is positive across all four ξ benchmarks, ranging from $+4.8\%$ ($\xi = 1.91$) to $+5.6\%$ ($\xi = 0.60$) per three-year period, although the estimate for any single period is imprecise. Appendix D2 reports the period-by-period values (Figure D1 and Table D1).

Three lines of evidence point to the same conclusion. Per-trip WTP remained stable over two decades despite rising demand and consumption, and the simplest explanation is that

[2] In 2012–2014, the $\xi = 0.86$ projection exceeds the upper bound of the 95% confidence interval by \$0.01.

Figure 4: Test of the Relative Price Change Framework: Observed and Projected WTP per Trip, 2003–2023



With no change in recreation-enabling assets ($g_E = 0$), the RPC projections for $\xi \in \{0.60, 0.74, 0.86\}$ track observed WTP through 2018–2020 but all four projections exceed its 95% confidence interval in 2021–2023. The $\xi = 1.91$ projection falls outside the intervals from 2012–2014 onward. The solid black line is the observed per-trip WTP ($\overline{\text{WTP}}_t$) with 95% confidence intervals. Dashed colored lines are projected paths from Equation 7, compounded forward from the 2003–2005 estimate, for four income-elasticity benchmarks from Drupp et al. (2025). g_C is real personal consumption expenditures per capita (Appendix D3 uses real GDP per capita instead). Each estimate is plotted at the midpoint year of its three-year period. Real 2023 USD.

new supply has kept pace. This conclusion is supported by the fact that city park acreage and total recreation trips grew at similar average rates (9.8% and 10.7% per three-year period, respectively). It is also supported by the evidence from inverting the RPC framework to back out the implied asset growth, which provides a direct estimate of how the enabling assets have changed. That exercise yields average implied growth rates that are positive across the four elasticities, providing no evidence that the enabling assets for local outdoor recreation have become scarcer. Together, these results indicate that the dispersed parks, trails, and public spaces enabling everyday outdoor recreation have, in aggregate, been sustainably maintained at the national level over the past two decades. It follows that the rising national value of local recreation reflects Americans using these assets more, not their growing scarcity.

6 Discussion

We provide a repeated, national-scale valuation of local outdoor recreation, spanning two decades of the ATUS. Three features distinguish these estimates from the existing recreation-valuation literature: they are national rather than site-specific, repeated rather than single-survey, and focused on ordinary neighborhood outdoor recreation trips that conventional

travel-cost studies rarely evaluate. Our WTP estimates for local recreation trips are easily replicable and could be updated with new years of ATUS data to understand trends in the ecosystem service value from local recreation in the United States. Tracking the value of ecosystem services over time at this scale is critical for evaluating whether their enabling assets are being managed sustainably (Fenichel, 2024).

Our per-trip WTP of $5.6 is low relative to the recreation valuation literature, as expected given the object of study. The Recreation Use Values Database, which compiles 4,023 consumer surplus estimates from 527 North American studies, reports a mean of $88 per person per activity day (2025 USD) (Oregon State University, College of Forestry, 2025). Even its lowest-valued categories, such as camping ($29), backpacking ($34), and picnicking ($43), exceed our per-trip estimate. The difference between prior estimates and this paper's is most likely explained by the unit of analysis and the activity in question. An activity day typically bundles a longer outing to a destination. In contrast, the average local trip in our data requires about 12 minutes of round-trip travel and just under an hour on site. Additionally, prior literature has focused on destination pursuits. Freshwater fishing, hunting, and wildlife viewing account for roughly 60 percent of the database's estimates, while walking, our most frequent activity, is not a primary category considered. The trips that dominate Americans' outdoor time are lower-valued per occasion and largely absent from prior literature, but are estimated to provide an average annual benefit of $93 billion for the 2021 to 2023 period.

It is useful to consider what our approach shares with conventional travel cost models that value a single or small set of destinations and where we depart due to valuing a diffuse set of sites. The core methods mirror the standard. We estimate a random utility model of trip-taking (McFadden, 1974a), treat the travel cost of reaching a recreation site as the price of a trip (Hotelling, 1947; Mäler, 1974), value travel time at the conventional one-third of the wage (Lupi et al., 2020), identify the price response from cross-sectional variation in that cost (Haab and McConnell, 2002), and convert our estimated parameters into consumer surplus (Parsons, 2017; Phaneuf and Requate, 2017b). As in any travel cost study, weak complementarity links trip-taking to the value of the asset, natural and built, that hosts it, and the resulting estimate is a use value. Our approach differs in the following three ways. First, the good is defined by where an activity takes place (outdoors, away from home) rather than by a named site. Because local parks, sidewalks, and schoolyards are close substitutes found in nearly every neighborhood, the relevant object of valuation is access to this network rather than to any single site. Second, because there is no single destination from which to compute distance, we observe travel time directly in the diary rather than constructing it from assumed distances and travel speeds. Third, for the same reason, we cannot compute

a non-recreator's distance to a site, so we impute her travel cost from similar recreators who face a comparable set of recreation opportunities.

We find multiple lines of evidence supporting the conclusion that accessible local outdoor spaces have been sustainably managed at the national level, despite large increases in the number of local outdoor recreation trips. The combination of a stable per-trip WTP and a 63 percent rise in national benefit suggests that recreation-enabling natural and built capital has not become more scarce per capita, even as it now provides substantially more aggregate benefit.

The RPC framework, proposed by Drupp et al. (2024, 2025) to update stale WTP estimates, also supports concluding that the enabling space for local outdoor recreation has been sustainably managed between 2003 and 2023. To show this result, we first test the RPC framework by projecting a single WTP estimate from our series forward, using income elasticities, ξ, from the literature and $g_E = 0$, which represents an unchanged stock of enabling assets. The projected path of WTP tracks our series of estimates through 2018–2020 for three of the four ξ benchmarks, but every projection overshoots our 2021–2023 estimate because per-trip WTP remained stable while consumption grew. We then use the full series of our WTP estimates and invert the RPC Equation 5 to calculate the growth rate of ecosystem services, g_E. We find that g_E is positive, about 5 percent per three-year period, implying that the assets enabling local recreation kept pace with consumption growth rather than becoming scarcer from 2003 through 2023. An alternative reading of the same evidence is that the income elasticity of WTP for local recreation is lower than the published benchmarks. Local recreation is an everyday good provided publicly at no charge, so WTP for it may respond little to rising income. Our data cannot separate a low ξ from a growing asset stock, because both imply a per-trip price that does not rise with consumption. Under either reading, the price of access to local outdoor space did not rise, which is not consistent with increasing scarcity.

Two non-mutually exclusive interpretations can explain how local recreation has not become scarcer despite large increases in the number of trips taken. First, the stock of recreation-enabling assets may have grown directly with demand. Second, local recreation trips may be sufficiently non-rival that parks, trails, and green spaces can accommodate additional visits without meaningfully diminishing each visitor's experience. In other words, a relatively static stock of local parks and similar outdoor spaces can accommodate greater use without increasing scarcity. The descriptive data favor the first explanation. As participation and aggregate benefits rose, the average round-trip travel time for a local recreation trip fell from 13.6 minutes in 2003–2005 to 10.5 minutes in 2021–2023 (Figure 1, Panel B). Travel times drifted only slightly downward from 2004 to 2019 (annual averages between

11.5 and 14.0 minutes), and the sharpest drop came in 2020 (8.5 minutes), which may partly reflect pandemic-era shifts toward recreating closer to home. Travel times had only partly recovered by 2023 (11.0 minutes) and showed no upward trend despite a 49 percent increase in trips per person. The decline is found across income quintiles and racial groups (Appendix Figure A2), and thus is unlikely to reflect a shift in which groups recreate. If sorting preferences remain constant over the same period, rising demand would increase travel times as new recreators visit a fixed or declining set of sites. Falling travel times, combined with rising participation, are consistent with accessible local recreation sites growing. Even without a national inventory of accessible local outdoor space, the stable WTP, stark rise in participation, and falling travel time tell a coherent story of the "accessible outdoors" being sustainably maintained, and possibly expanded, at the national scale.

Our conclusions are subject to three limitations. First, our claim that the enabling assets have not become scarcer depends on the g_E inferred through the RPC framework rather than on an independent census of recreation land. We corroborate it with acreage estimates from The Trust for Public Land (2025), and support our conclusion of sustainable management with descriptive evidence, but we do not observe the stock of outdoor recreation sites directly. Second, travel times are imputed for non-recreators through matching. Our estimates are stable in the number of matched recreators at the caliper selected for balance, but they are sensitive to the caliper (Appendix C3), and the imputation is model-based. Finally, because we condition on activity type but not on site quality, "no scarcer" refers to the availability of access and does not account for the quality of the land, water, or air where a recreation trip takes place. The quality of location that supports local recreation is an important distinction in natural-capital accounting, but one that this paper does not address.

How the quantity demanded for a service changes over time also matters for projecting its value, and our 20-year series shows that ignoring this change may lead to the largest errors. Because per-trip WTP remained stable, the 63 percent increase in the national annual value of local recreation is primarily explained by more trips per person (44 to 65 per year) and a growing population. Even if a benefit-cost analysis were to update a previously estimated WTP with a method like the RPC framework, estimates of aggregate benefit growth would be significantly understated if participation changes are ignored. Accurately projecting the future value of ecosystem services is sensitive to how the quantity demanded evolves, not just to changes in the WTP.

When the question changes from estimating the benefit of local outdoor recreation to who benefits, the choice of how to price travel time matters. The two pricing methods used in Figure 3 map onto two distinct policy questions. Financing new parks and trails is a function of the ability to pay. Investments in parks and green space are funded through taxes and

donations, so a WTP measure that embeds the wage distribution is appropriate for questions of how to finance expanded recreation infrastructure. Measuring the non-market benefit that outdoor spaces deliver across demographic groups is a different question, however, and one in which embedding the wage distribution may understate the realized benefits that frequent, low-income users derive from accessing free outdoor recreation opportunities (Ando et al., 2024).

Under one-third-wage pricing for the opportunity cost of time, the top income quintile's WTP per trip is 13 times that of the bottom quintile. However, the top quintile takes only about 1.6 times as many trips per year. The WTP gap is an order of magnitude larger than the participation gap. The replacement wage compresses this gap considerably (Figure 3), revealing a pattern more consistent with the descriptive statistics in Section 2: everyone recreates, and the public-good nature of the outdoors delivers non-market benefits across the income distribution. Presenting the two pricing methods side by side, rather than choosing between them, gives a more complete picture of who can finance local recreation opportunities versus who benefits from them.

We estimate the value of a category of public capital that has largely been missing from the recreation valuation literature and the national accounts. The places that generate most of the value U.S. residents derive from outdoor recreation may not be the "grand scenic wonders" that travel cost models were built to evaluate (Krutilla, 1967). They are places that support frequent, everyday outdoor time, such as corner parks, sidewalks, greenways, and schoolyards that function as neighborhood parks, regardless of their designation. These are assets missing from U.S. land accounts (Wentland et al., 2020) as well as national accounts because they are publicly provided and free at the point of use, so no market transaction records their contribution. Our estimates suggest these local outdoor recreation sites currently deliver roughly $93 billion in annual recreational use value, a lower bound on their total value, and that they have not become scarcer over two decades.

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A Characterizing Local Recreation

Table A1: Outdoor Activity Categories

Rank	Activity Category	Share of Trips (%)	Avg. Annual Hours (per U.S. Res.)	Total U.S. Hours per Year (bn)	Avg. Time on Site per Trip (min)	Avg. Travel Time per Trip (min)
1	Playing with Pets	36.5	10.77	2.665	33	10.2
2	Walking	22.9	11.11	2.751	54	10.0
3	Playing with Children	10.7	4.58	1.135	49	22.9
4	Socializing and Relaxing	9.5	7.62	1.887	85	36.2
5	Running	6.3	3.02	0.747	53	11.2
6	Sports	3.7	4.63	1.145	141	38.2
7	Biking	3.1	2.36	0.583	86	12.1
8	Fishing and Hunting	2.1	4.88	1.208	267	53.0
9	Water sports	1.7	2.46	0.610	168	37.6

The nine most popular outdoor activity categories (aggregated from six-digit ATUS codes; the residual "Other" category is not shown), sorted by share of all outdoor recreation trips. For each category, we report its share of trips, average annual hours per U.S. resident, total annual hours nationally (billions), average time on site per trip (in minutes), and average round-trip travel time per trip (in minutes). A trip is assigned to the category of its longest activity; annual hours count all time spent in the category, including time within trips whose longest activity falls in another category. Activities are limited to those occurring outdoors (not at home) during non-work leisure hours. All estimates use ATUS sample weights for the 2003–2023 period. All outdoor recreation trips are included, not only local trips. Everyday outdoor activities such as playing with pets or children, walking, and socializing and relaxing dominate, while the pursuits that recreation demand studies typically focus on (running, biking, fishing) rank lower.

Table A2: Trips, Annual Hours, Time on Site, and Travel Time for All Outdoor Activities

Rank	Six-Digit Activity Description	Activity Category	Share of Trips (%)	Avg. Annual Hours (per U.S. Res.)	Total U.S. Hours per Year (bn)	Avg. Time on Site per Trip (min)	Avg. Travel Time per Trip (min)
1	Walking	Walking	22.9	11.11	2.751	54	10.0
2	Walking, exercising, playing with animals (2008+)	Playing with Pets	30.1	8.80	2.179	32	9.9
3	Socializing and communicating with others	Socializing and Relaxing	4.2	3.44	0.851	93	29.8
4	Running	Running	6.3	2.99	0.739	53	11.0
5	Fishing	Fishing and Hunting	1.3	2.83	0.700	257	50.6
6	Biking	Biking	3.1	2.33	0.578	85	11.9
7	Hunting	Fishing and Hunting	0.8	2.02	0.499	285	57.6
8	Relaxing, thinking	Socializing and Relaxing	1.9	1.55	0.384	89	38.0
9	Eating and drinking	Socializing and Relaxing	2.8	1.53	0.380	44	42.8
10	Golfing	Other	0.8	1.51	0.374	217	36.5
11	Care for animals and pets (not veterinary care)	Playing with Pets	4.7	1.45	0.360	34	10.7
12	Participating in water sports	Water sports	1.0	1.31	0.324	155	33.2
13	Hiking	Other	0.9	1.17	0.290	150	51.6

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Rank	Six-Digit Activity Description	Activity Category	Share of Trips (%)	Avg. Annual Hours (per U.S. Res.)	Total U.S. Hours per Year (bn)	Avg. Time on Site per Trip (min)	Avg. Travel Time per Trip (min)
14	Playing with hh children, not sports	Playing with Children	1.3	1.17	0.289	99	25.0
15	Boating	Water sports	0.7	1.09	0.270	184	39.9
16	Attending hh children's events	Playing with Children	1.0	1.07	0.264	120	36.4
17	Attending or hosting par-ties/receptions/ceremoni	Socializing and Relaxing	0.4	0.84	0.209	218	47.8
18	Looking after hh children (as a primary activity)	Playing with Children	1.0	0.81	0.202	97	25.9
19	Playing basketball	Sports	0.5	0.62	0.154	149	19.9
20	Playing sports n.e.c.	Sports	0.5	0.58	0.145	125	33.6
21	Watching baseball	Sports	0.3	0.50	0.123	185	65.0
22	Playing soccer	Sports	0.4	0.45	0.111	128	42.1
23	Playing sports with hh children	Playing with Children	0.8	0.40	0.098	52	8.4
24	Care for animals and pets (not veterinary care) (2008+)	Playing with Pets	1.3	0.39	0.097	34	14.6
25	Skiing, ice skating, snowboarding	Other	0.2	0.37	0.092	180	66.1
26	Playing racquet sports	Sports	0.3	0.35	0.087	132	26.6
27	Watching football	Sports	0.2	0.33	0.082	176	47.1
28	Waiting for/with hh children	Playing with Children	2.0	0.29	0.072	16	13.6
29	Watching soccer	Sports	0.2	0.27	0.067	130	41.3

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Rank	Six-Digit Activity Description	Activity Category	Share of Trips (%)	Avg. Annual Hours (per U.S. Res.)	Total U.S. Hours per Year (bn)	Avg. Time on Site per Trip (min)	Avg. Travel Time per Trip (min)
30	Attending religious services	Other	0.2	0.26	0.064	149	48.5
31	Attending nonhh children's events	Playing with Children	0.3	0.25	0.063	112	35.8
32	Playing softball	Sports	0.2	0.25	0.062	144	42.9
33	Playing games	Sports	0.2	0.24	0.059	141	22.8
34	Rollerblading	Other	0.3	0.24	0.059	104	14.4
35	Playing baseball	Sports	0.2	0.22	0.054	161	42.7
36	Picking up/dropping off hh children	Playing with Children	3.0	0.20	0.050	7	24.4
37	Participating in equestrian sports	Other	0.1	0.17	0.043	149	24.6
38	Playing volleyball	Sports	0.1	0.14	0.036	194	38.7
39	Playing football	Sports	0.1	0.14	0.035	137	24.8
40	Attending sporting events, n.e.c.	Sports	0.1	0.14	0.034	145	40.2
41	Watching vehicle touring/racing	Other	0.0	0.13	0.033	321	76.5
42	Looking after nonhh children (as primary activity)	Playing with Children	0.2	0.13	0.032	91	23.9
43	Watching softball	Sports	0.1	0.13	0.032	160	35.6
44	Attending meetings for personal interest (not volunteering)	Socializing and Relaxing	0.1	0.13	0.032	183	38.9
45	Working out, unspecified	Sports	0.2	0.12	0.029	62	58.3

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Rank	Six-Digit Activity Description	Activity Category	Share of Trips (%)	Avg. Annual Hours (per U.S. Res.)	Total U.S. Hours per Year (bn)	Avg. Time on Site per Trip (min)	Avg. Travel Time per Trip (min)
46	Participation in religious practices	Other	0.3	0.11	0.027	43	38.6
47	Animal and pet care assistance for nonhh adults	Playing with Pets	0.3	0.11	0.026	36	12.2
48	Climbing, spelunking, caving	Other	0.0	0.09	0.022	280	73.7
49	Relaxing and leisure, n.e.c.	Socializing and Relaxing	0.1	0.08	0.020	139	30.4
50	Helping nonhh adults, n.e.c.	Other	0.1	0.07	0.018	104	28.2
51	Waiting associated with helping hh adults	Other	0.3	0.06	0.016	23	36.3
52	Playing sports with nonhh children	Playing with Children	0.1	0.06	0.016	65	15.5
53	Watching basketball	Sports	0.1	0.05	0.013	121	25.1
54	Watching volleyball	Sports	0.0	0.05	0.011	182	77.4
55	Physical care for hh children	Playing with Children	0.2	0.05	0.011	20	24.2
56	Watching water sports	Water sports	0.0	0.04	0.011	260	125.4
57	Caring for and helping nonhh children, n.e.c.	Playing with Children	0.0	0.03	0.009	165	63.9
58	Helping hh adults	Other	0.0	0.03	0.008	86	38.4
59	Watching running	Running	0.0	0.03	0.008	143	48.0
60	Dropping off/picking up nonhh children	Playing with Children	0.5	0.03	0.008	7	26.3

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Rank	Six-Digit Activity Description	Activity Category	Share of Trips (%)	Avg. Annual Hours (per U.S. Res.)	Total U.S. Hours per Year (bn)	Avg. Time on Site per Trip (min)	Avg. Travel Time per Trip (min)
61	Watching fishing	Fishing and Hunting	0.0	0.03	0.008	166	25.9
62	Waiting for/with nonhh children	Playing with Children	0.2	0.03	0.008	16	24.4
63	Playing rugby	Other	0.0	0.03	0.007	232	33.1
64	Watching equestrian sports	Other	0.0	0.03	0.007	189	52.2
65	Watching rodeo competitions	Other	0.0	0.02	0.006	171	50.3
66	Watching racquet sports	Sports	0.0	0.02	0.006	277	127.4
67	Watching biking	Biking	0.0	0.02	0.006	140	42.2
68	Physical care for nonhh children	Playing with Children	0.0	0.02	0.006	95	52.3
69	Religious education activities (2007+)	Other	0.0	0.02	0.005	170	20.5
70	Caring for and helping hh children, n.e.c.	Playing with Children	0.0	0.02	0.005	55	14.4
71	Watching boating	Water sports	0.0	0.02	0.005	190	83.6
72	Attending/hosting social events, n.e.c.	Socializing and Relaxing	0.0	0.02	0.004	114	15.2
73	Playing hockey	Other	0.0	0.02	0.004	119	51.5
74	Extracurricular club activities	Other	0.0	0.02	0.004	240	25.0
75	Watching golfing	Other	0.0	0.02	0.004	413	66.9
76	Pet and animal care, n.e.c.	Playing with Pets	0.0	0.01	0.004	72	57.1

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Rank	Six-Digit Activity Description	Activity Category	Share of Trips (%)	Avg. Annual Hours (per U.S. Res.)	Total U.S. Hours per Year (bn)	Avg. Time on Site per Trip (min)	Avg. Travel Time per Trip (min)
77	Socializing, relaxing, and leisure, n.e.c.	Socializing and Relaxing	0.0	0.01	0.003	83	43.2
78	Socializing, relaxing, and leisure as part of job	Socializing and Relaxing	0.0	0.01	0.003	141	35.0
79	Watching hockey	Other	0.0	0.01	0.002	123	40.1
80	Sports and exercise as part of job	Sports	0.0	0.01	0.002	47	12.6
81	Helping household adults, n.e.c.	Other	0.0	0.01	0.002	56	34.0
82	Looking after hh adult (as a primary activity)	Other	0.0	0.01	0.002	125	14.4
83	Sports, exercise, and recreation, n.e.c.	Sports	0.0	0.01	0.002	132	25.3
84	Talking with/listening to nonhh children	Playing with Children	0.0	0.01	0.001	28	19.6
85	Physical care for nonhh adults	Other	0.0	0.01	0.001	19	39.8
86	Watching skiing, ice skating, snowboarding	Other	0.0	0.00	0.001	43	14.6
87	Activities related to nonhh child's health, n.e.c.	Playing with Children	0.0	0.00	0.001	90	60.0
88	Watching rugby	Other	0.0	0.00	0.001	80	14.0
89	Participating in rodeo competitions	Other	0.0	0.00	0.001	450	180.0

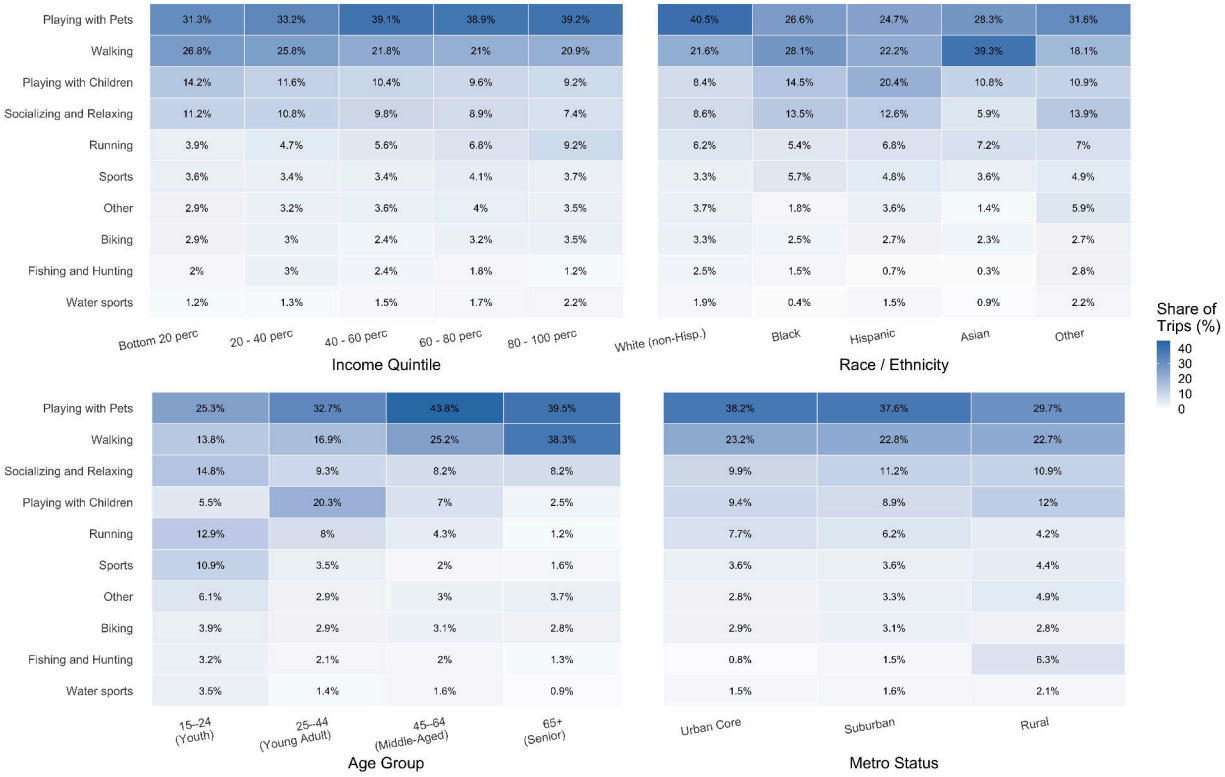
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Rank	Six-Digit Activity Description	Activity Category	Share of Trips (%)	Avg. Annual Hours (per U.S. Res.)	Total U.S. Hours per Year (bn)	Avg. Time on Site per Trip (min)	Avg. Travel Time per Trip (min)
90	Religious and spiritual activities, n.e.c.	Other	0.0	0.00	0.001	179	8.4
91	Caring for and helping nonhh members, n.e.c.	Other	0.0	0.00	0.000	–	–
92	Caring for household adults, n.e.c.	Other	0.0	0.00	0.000	15	0.0
93	Waiting assoc w/religious and spiritual activities	Other	0.0	0.00	0.000	16	17.3
94	Physical care for hh adults	Other	0.0	0.00	0.000	11	10.4
95	Looking after nonhh adult (as a primary activity)	Other	0.0	0.00	0.000	9	28.6
96	Watching rollerblading	Other	0.0	0.00	0.000	15	40.0
97	Caring for and helping hh members, n.e.c.	Other	0.0	0.00	0.000	3	25.0

Each six-digit ATUS outdoor activity, its aggregated activity category, share of trips, average annual hours per U.S. resident, total annual hours nationally (billions), average time on site per trip (in minutes), and average round-trip travel time per trip (in minutes). Column definitions match Table A1: a trip is assigned to the activity of its longest episode, and annual hours count all time spent in the activity. Activities that are never the longest activity of a trip have a zero trip share and no per-trip values (–). All outdoor recreation trips are included—not restricted to local trips (round-trip travel time $\leq$ 60 minutes). Travel time is averaged across trips (weighted by ATUS sample weights), not summed over the year.

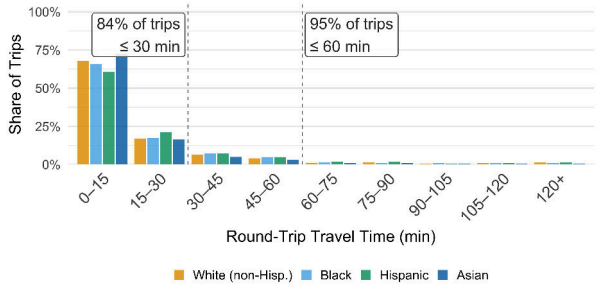
Figure A1: Share of Trips Taken, by Demographic Group



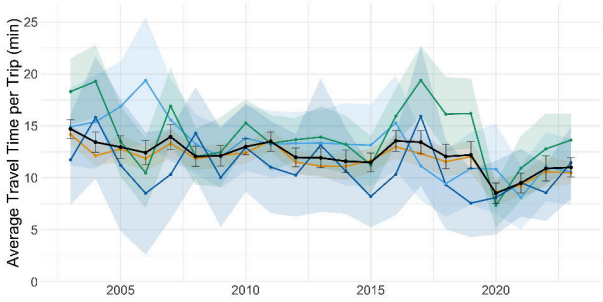
Weighted share of all outdoor recreation trips by activity type, displayed by demographic groups. ATUS 2003–2023. All outdoor trips are included, not only local trips. Darker shading indicates a larger share; the color scale is common to all four panels.

Figure A2: Local Outdoor Recreation by Race/Ethnicity

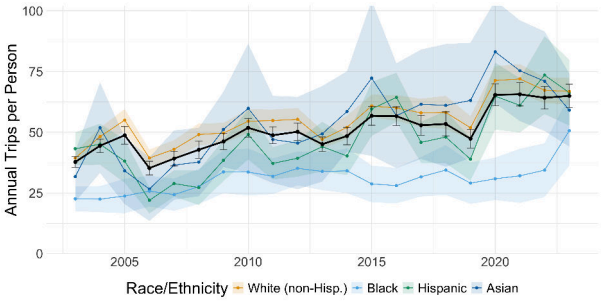
(A) Distribution of Round-Trip Travel Time



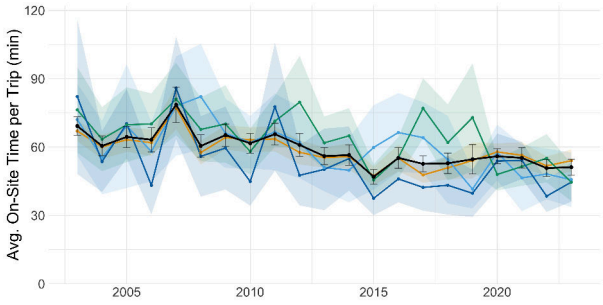
(B) Average Round-Trip Travel Time



(C) Annual Local Recreation Trips per Person



(D) Average On-Site Time per Trip



Panel (A) shows the share of each group's outdoor recreation trips (local and non-local) in each round-trip travel-time bin; bins include their upper bound, and the dashed lines mark the 30- and 60-minute thresholds. Panels (B)–(D) describe local outdoor recreation only (round-trip travel time $\leq$ 60 minutes). Panel (B) plots the annual weighted-average round-trip travel time by race/ethnicity; the black line is the national average across all individuals. Panel (C) shows annual trips per U.S. resident by race/ethnicity (denominator is the full population, including non-recreators), with the black line denoting the national average. Panel (D) plots the annual weighted-average time spent at the recreation activity per trip (dwell time) by race/ethnicity, with the black line denoting the national average. All estimates use ATUS sample weights for 2003–2023; standard errors in (B)–(D) are computed using the ATUS replicate-weight methodology (160 replicates).

B Travel Cost Model

This section defines the variables that enter the travel cost model and reports specification selection and robustness analyses.

B1 Variable Definitions and Opportunity Cost of Time

The conditional indirect utility of taking a trip ($j = 1$) in time period t is given by Equation 1 in the main text, where β_0 is the baseline utility of taking a trip and π_i is the travel cost of a trip, in which travel time is converted to a monetary cost using one-third of individual i's average hourly wage. The travel cost serves as the price of a recreation trip, since local outdoor recreation is typically free at the point of use. For our main analyses, we follow the literature and assume the opportunity cost of time is one-third of an individual's wage rate (Lupi et al., 2020). We present a secondary analysis where the opportunity cost of an hour is valued at $16.08, the wage of a general-purpose domestic worker, which the Bureau of Economic Analysis uses as the replacement wage to value household production (Bureau of Economic Analysis, 2018).

$\textbf{State}_i$ is a vector of state fixed effects. For individuals who take a recreation trip, $\boldsymbol{D}$ is a vector proxying for trip-quality controls. It measures the type of activity an individual participated in. There are nine categories (playing with pets, socializing and relaxing, playing with children, running, fishing and hunting, water sports, biking, sports, and other), with walking as the omitted reference. For individuals who do not take a trip, $\boldsymbol{D}$ measures the expected probability of participating in each category of outdoor activity. y_i is family income measured in 2023 USD, taken as the midpoint of the income bin the respondent reports in the ATUS. It is used as a control and as the basis for the opportunity cost of travel time: we compute the average hourly wage as family income divided by 2,080 hours. $\boldsymbol{X}_i$ is a vector of demographic characteristics: race/ethnicity indicators (non-Hispanic White omitted), age in years, an indicator for living in a non-metropolitan area, and a sex indicator. Car_i is an indicator for whether the respondent traveled by car on the diary day. L_i^{eat} is the average travel time in the respondent's county-year for eating or drinking out, which is included as a potential indoor substitute activity to outdoor recreation. $G_{r,m}$ is the monthly real retail price of gasoline in the Petroleum Administration for Defense District (PADD) r containing the respondent's state of residence in diary month m, in 2023 USD per gallon (construction described below).

We normalize the conditional indirect utility of not taking a trip to zero, $V^0 = 0$. Any constant would be arbitrary when estimating our travel cost model, as it is absorbed by β_0, but choosing zero simplifies the calculation of consumer surplus in later steps.

B2 Model Selection and Robustness Across Controls

This subsection documents how the travel-cost specification in Equation 1 was selected by progressively adding controls, and demonstrates that the estimated WTP per trip is robust to those choices. Models M1–M11 build up to the preferred specification as follows: M1 includes only travel cost and state fixed effects; M2 adds the trip-quality controls $\boldsymbol{D}_i$; M3 adds family income y_i; M4 adds race/ethnicity; M5 adds age; M6 adds the rural indicator; M7 adds the car-travel indicator; M8 adds the out-of-home leisure substitute control L_i^{eat}; M9 adds the sex indicator; M10 adds the regional real price of gasoline $G_{r,m}$; and M11 adds the interaction of gasoline price with the car-travel indicator. M11 is our preferred specification and the model presented in the main text. M12 is a robustness model that additionally interacts gasoline prices with travel costs, as discussed below.

To construct the gasoline price variable, we pull monthly retail prices (dollars per gallon) for each of the five Petroleum Administration for Defense Districts (PADDs), which are the U.S. Energy Information Administration's (EIA) regional groupings of states for reporting petroleum statistics (East Coast, Midwest, Gulf Coast, Rocky Mountain, and West Coast), from the EIA (U.S. Energy Information Administration, 2025). We convert these prices to real 2023 USD using the same annual CPI series that adjusts family income for inflation. We map each respondent to the PADD region containing her state of residence using a state-to-PADD crosswalk and merge the monthly price onto each respondent by the year and month of her diary day. The merged price captures the monetary cost of driving at the time and place of each diary, so the travel-cost coefficient in M10–M12 is identified holding this cost fixed.

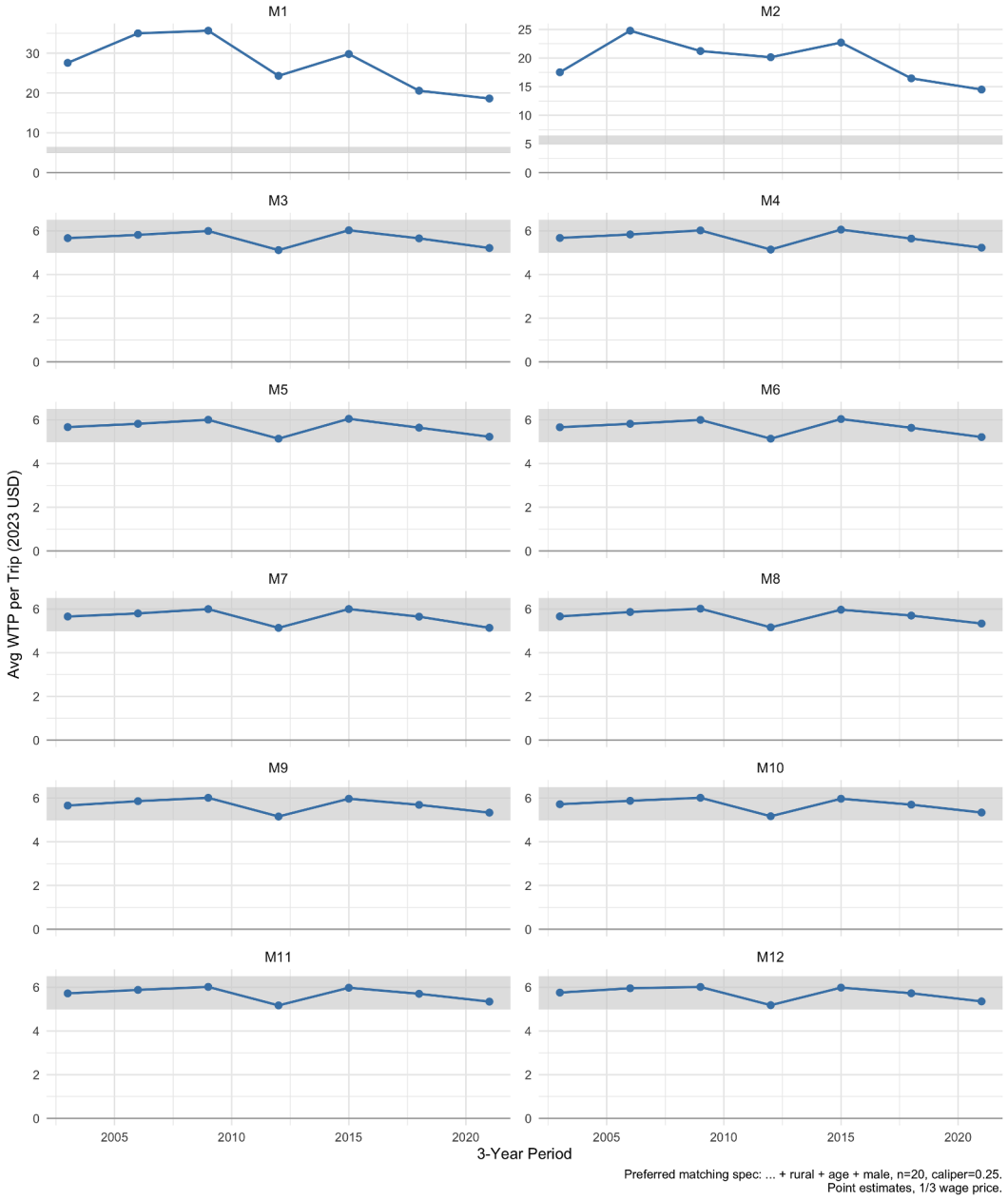
Table B1 reports M1–M12 for the WTP travel-cost model, using one-third the wage rate as the opportunity cost of time. The travel-cost coefficient $\hat{\beta}_1$ stabilizes by M3: once trip quality and income are controlled for, the estimated price response is insensitive to further observable controls. This stability does not rule out residual sorting on unobservables, which we discuss in Section 3.4. The coefficient is unchanged as sex and the gasoline terms are added in M9–M11, and per-trip WTP is stable across these models, ranging from \$5.59 to \$5.65 in M3 through M11 and equal to \$5.65 in the preferred specification.

M12 allows the monetary and time costs of a trip to interact, so that the marginal disutility of travel cost varies with the price of gasoline. Two features of the interacted specification require care when comparing it to the preferred model. First, in M12, the coefficient on travel cost is the marginal effect evaluated at a gasoline price of zero, which is outside the data's support. The relevant comparison to the preferred model's $\hat{\beta}_1$ is the average marginal effect of travel cost, $\hat{\beta}_1 + \hat{\gamma}\,\bar{G}$, where $\hat{\gamma}$ is the interaction coefficient and $\bar{G}$ is the mean gasoline price in the estimation sample (\$3.64 per gallon). The average marginal

effect in M12 is within a fraction of a percent of the preferred model's travel-cost coefficient and statistically indistinguishable from it. Second, because the marginal utility of money is individual-specific in the interacted model, converting the log-sum of utilities into dollars requires the individual-specific conversion $-1/(\hat{\beta}_1 + \hat{\gamma}\, G_i)$ rather than the constant $-1/\hat{\beta}_1$. With this conversion, the average WTP per trip in M12 differs from the preferred model by less than one percent. Although the interaction coefficient is statistically different from zero, allowing it does not change the welfare estimates: WTP per trip is \$5.67 in M12 and \$5.65 in M11. We therefore use the simpler specification (M11) for all welfare estimates; using M12 instead would not materially change any estimate in the paper.

Figure B1 shows the time-series counterpart: the average value per trip across the seven 3-year periods, separately for each model M1–M12, with the matching specification set to the preferred choice. Again, we see that for Models 3 through 12, WTP per trip estimates are steady between \$5.00 and \$6.50.

Figure B1: Average Value per Trip over Time, by Regression Model (M1–M12, Preferred Matching)



Average value per trip over the seven 3-year periods, by regression model M1–M12. Preferred matching specification (time uses + holiday/weekend + rural, age, sex; $n = 20$, caliper $= 0.25$). Point estimates: travel time priced at one-third of the wage. The grey band marks \$5.00–\$6.50.

Table B1: Logit Travel-Cost Model Buildup, M1–M12 (1/3 Wage, All Years Pooled, Preferred Matching Specification)

Variable	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12
Dependent Var.:	trip	trip	trip	trip	trip	trip	trip	trip	trip	trip	trip	trip
Constant	-2.071***	-3.570***	-3.916***	-3.691***	-3.885***	-3.852***	-3.865***	-3.899***	-3.901***	-4.040***	-3.947***	-3.869***
	(0.0636)	(0.0654)	(0.0691)	(0.0698)	(0.0725)	(0.0732)	(0.0751)	(0.0822)	(0.0824)	(0.0872)	(0.1082)	(0.1183)
Travel cost	-0.0434***	-0.0605***	-0.2049***	-0.2047***	-0.2045***	-0.2047***	-0.2049***	-0.2034***	-0.2034***	-0.2033***	-0.2033***	-0.2413***
	(0.0030)	(0.0029)	(0.0068)	(0.0068)	(0.0068)	(0.0068)	(0.0068)	(0.0068)	(0.0068)	(0.0068)	(0.0068)	(0.0205)
Playing with pets		3.278***	3.059***	3.022***	3.021***	3.014***	3.013***	3.021***	3.021***	3.021***	3.021***	3.021***
		(0.0670)	(0.0728)	(0.0722)	(0.0723)	(0.0725)	(0.0725)	(0.0730)	(0.0729)	(0.0730)	(0.0730)	(0.0729)
Socializing and relaxing		2.832***	3.170***	3.163***	3.181***	3.178***	3.177***	3.176***	3.176***	3.177***	3.178***	3.178***
		(0.0958)	(0.0935)	(0.0932)	(0.0931)	(0.0932)	(0.0932)	(0.0941)	(0.0941)	(0.0940)	(0.0940)	(0.0940)
Playing with children		3.516***	3.652***	3.655***	3.694***	3.690***	3.688***	3.703***	3.703***	3.701***	3.702***	3.702***
		(0.0732)	(0.0775)	(0.0775)	(0.0773)	(0.0774)	(0.0774)	(0.0778)	(0.0778)	(0.0778)	(0.0778)	(0.0778)
Running		3.307***	2.696***	2.688***	2.726***	2.719***	2.717***	2.735***	2.735***	2.731***	2.731***	2.732***
		(0.0866)	(0.1089)	(0.1082)	(0.1081)	(0.1083)	(0.1083)	(0.1085)	(0.1086)	(0.1086)	(0.1086)	(0.1086)
Fishing and hunting		3.984***	4.585***	4.536***	4.572***	4.587***	4.585***	4.592***	4.590***	4.591***	4.591***	4.597***
		(0.1086)	(0.1137)	(0.1139)	(0.1138)	(0.1137)	(0.1138)	(0.1139)	(0.1141)	(0.1142)	(0.1142)	(0.1143)
Water sports		3.452***	3.493***	3.443***	3.477***	3.472***	3.471***	3.482***	3.482***	3.474***	3.474***	3.474***
		(0.1370)	(0.1566)	(0.1554)	(0.1557)	(0.1560)	(0.1560)	(0.1571)	(0.1571)	(0.1573)	(0.1573)	(0.1572)
Biking		3.391***	3.046***	3.020***	3.036***	3.029***	3.029***	3.021***	3.020***	3.016***	3.017***	3.019***
		(0.0995)	(0.1179)	(0.1175)	(0.1175)	(0.1177)	(0.1177)	(0.1199)	(0.1199)	(0.1199)	(0.1199)	(0.1199)
Ball or racquet sports		3.346***	3.684***	3.696***	3.747***	3.743***	3.742***	3.746***	3.745***	3.743***	3.743***	3.744***
		(0.1110)	(0.1167)	(0.1173)	(0.1173)	(0.1174)	(0.1174)	(0.1181)	(0.1181)	(0.1180)	(0.1180)	(0.1179)
Other activity		3.453***	3.794***	3.758***	3.779***	3.778***	3.776***	3.792***	3.792***	3.797***	3.797***	3.798***
		(0.1052)	(0.1141)	(0.1138)	(0.1137)	(0.1137)	(0.1137)	(0.1139)	(0.1139)	(0.1137)	(0.1137)	(0.1137)
Family income			1.01e-5***	9.3e-6***	9.5e-6***	9.45e-6***	9.44e-6***	9.4e-6***	9.4e-6***	9.42e-6***	9.42e-6***	9.4e-6***
			(2.51e-7)	(2.53e-7)	(2.54e-7)	(2.54e-7)	(2.56e-7)	(2.57e-7)	(2.57e-7)	(2.57e-7)	(2.57e-7)	(2.57e-7)
Race: Asian				-0.0993*	-0.0728	-0.0805	-0.0796	-0.0770	-0.0771	-0.0779	-0.0784	-0.0795
				(0.0427)	(0.0429)	(0.0429)	(0.0429)	(0.0431)	(0.0431)	(0.0430)	(0.0431)	(0.0430)
Race: Black				-0.5729***	-0.5648***	-0.5726***	-0.5717***	-0.5701***	-0.5701***	-0.5719***	-0.5722***	-0.5728***
				(0.0277)	(0.0277)	(0.0278)	(0.0277)	(0.0279)	(0.0279)	(0.0279)	(0.0279)	(0.0279)
Race: Hispanic				-0.2528***	-0.2223***	-0.2278***	-0.2273***	-0.2253***	-0.2254***	-0.2256***	-0.2259***	-0.2263***

Continued on next page...

Table B1 continued from previous page

Variable	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12
				(0.0240)	(0.0243)	(0.0244)	(0.0244)	(0.0245)	(0.0245)	(0.0245)	(0.0245)	(0.0245)
Race: Other				-0.1135*	-0.0865	-0.0873	-0.0868	-0.0860	-0.0861	-0.0861	-0.0860	-0.0874
				(0.0498)	(0.0499)	(0.0498)	(0.0498)	(0.0502)	(0.0502)	(0.0502)	(0.0502)	(0.0502)
Age					0.0033***	0.0033***	0.0034***	0.0033***	0.0033***	0.0033***	0.0033***	0.0033***
					(0.0004)	(0.0004)	(0.0004)	(0.0004)	(0.0004)	(0.0004)	(0.0004)	(0.0004)
Non-metropolitan						-0.0891***	-0.0893***	-0.0875***	-0.0876***	-0.0881***	-0.0881***	-0.0887***
						(0.0225)	(0.0226)	(0.0226)	(0.0226)	(0.0226)	(0.0226)	(0.0226)
Traveled by car							0.0161	0.0143	0.0143	0.0133	-0.1047	-0.0640
							(0.0188)	(0.0189)	(0.0189)	(0.0189)	(0.0829)	(0.0816)
Travel time to restaurants (county-year)								0.0014	0.0014	0.0014	0.0014	0.0014
								(0.0014)	(0.0014)	(0.0014)	(0.0014)	(0.0014)
Male									0.0037	0.0035	0.0036	0.0039
									(0.0152)	(0.0152)	(0.0152)	(0.0152)
Regional gasoline price (real)										0.0423***	0.0169	-0.0039
										(0.0090)	(0.0200)	(0.0238)
Traveled by car x Regional gasoline price (real)											0.0321	0.0210
											(0.0220)	(0.0216)
Travel cost x gasoline price												0.0104*
												(0.0052)
State FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
S.E.: Clustered	by: caseid	by: caseid	by: caseid	by: caseid	by: caseid	by: caseid	by: caseid	by: caseid	by: caseid	by: caseid	by: caseid	by: caseid
Observations	229,427	229,427	208,445	208,445	208,445	208,445	208,445	206,039	206,039	206,039	206,039	206,039
Squared Cor.	0.01235	0.24202	0.25795	0.25881	0.25767	0.25728	0.25731	0.25742	0.25741	0.25751	0.25753	0.25770
Pseudo R2	0.01163	0.10063	0.12048	0.12426	0.12468	0.12478	0.12479	0.12478	0.12478	0.12495	0.12496	0.12504
BIC	186,212.9	169,613.9	145,485.1	144,913.0	144,856.0	144,850.4	144,861.8	143,115.3	143,127.5	143,113.6	143,123.1	143,123.5

Notes: Logit estimates of trip-taking on travel cost and progressively added controls. Travel cost is priced at one-third of the individual's average hourly wage. Sample pooled across 2003–2023. Preferred matching specification (time uses + holiday/weekend + rural, age, sex; $n = 20$, caliper = 0.25). Standard errors clustered by respondent in parentheses. M1: travel cost + state FE; M2: + $\boldsymbol{D}_i$; M3: + income; M4: + race; M5: + age; M6: + rural; M7: + car; M8: + L_i^{eat}; M9: + sex; M10: + regional gas price; M11: + gas price × car (preferred); M12: + gas price × travel cost (appendix interaction check). Adding sex, the regional real gasoline price, and its interaction with car travel leaves per-trip WTP essentially unchanged ($5.64–$5.65 across M8–M11), and further interacting gasoline price with travel cost (M12) leaves the average travel-cost response and per-trip WTP unchanged.

Table B2: Pooled Value of Local Outdoor Recreation, 2003–2023

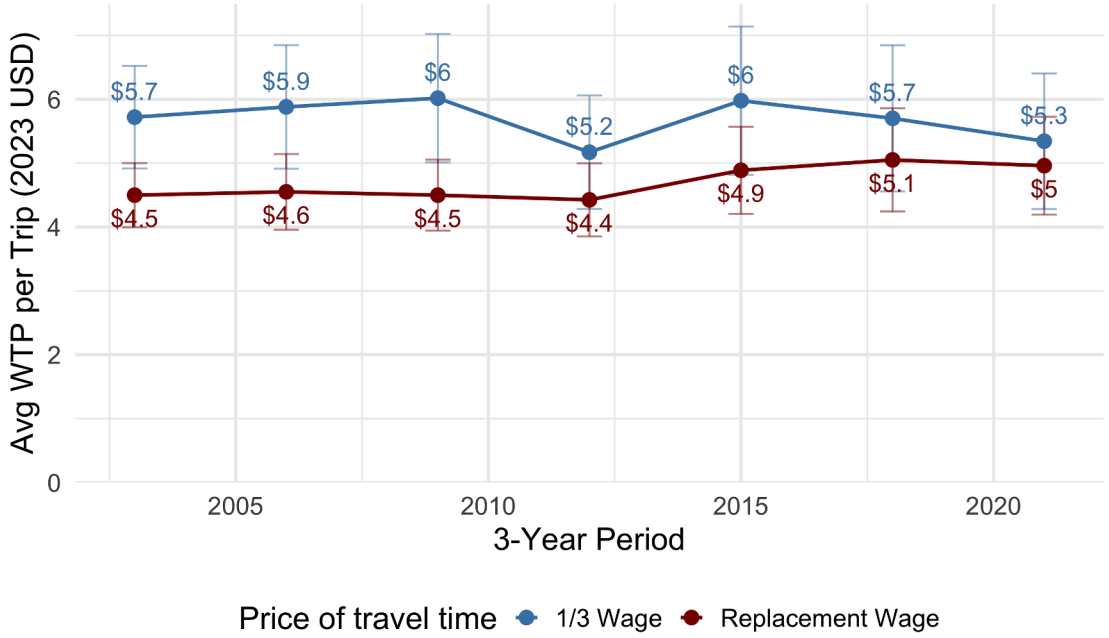
Quantity	Estimate	Std. Error
Avg WTP per local recreation trip ($)	5.65	0.18
Annual value per person ($)	288	11
Annual national value (Billion $)	71.7	2.7

Pooled estimates of the value of local outdoor recreation across the 20-year study period. Per-trip WTP is estimated by fitting the full travel cost model (Equation 1) on all years pooled, with delta-method standard errors. Annual per-person and national values are simple averages across the seven 3-year periods, with standard errors calculated as $\sqrt{\overline{SE_t^2}/T}$ treating periods as independent. Travel time is priced at one-third of the respondent's average hourly wage. All values are in real 2023 USD.

B3 Alternative Value of Travel Time

The main results price travel time at one-third of the respondent's average hourly wage, following the standard travel-cost convention. Here we re-estimate the full travel cost model (Equation 1) using a single price of $16.08 per hour applied to every respondent: the wage of a general-purpose domestic worker, which the Bureau of Economic Analysis uses as the replacement wage for household production. Figure B2 shows the period-by-period average WTP per trip under both pricing methods. Per-trip WTP is somewhat lower under the replacement wage, but the 95% confidence intervals of the two series overlap in all seven periods.

Figure B2: Trend in WTP for Different Values of Travel Time



Average WTP per local recreation trip by 3-year period under the two pricing methods. "1/3 Wage" prices travel time at one-third of the respondent's average hourly wage; "Replacement Wage" prices travel time at $16.08 per hour for every respondent. Both estimates use the preferred matching specification and the full travel cost model (Equation 1). Error bars are 95% confidence intervals from delta-method standard errors. All values are in real 2023 USD.

B4 Heterogeneous WTP by Demographic Subgroup

The subgroup estimates shown in Figure 3 of the main text are produced by re-fitting the full travel cost model (Equation 1) separately within each demographic subgroup, rather than computing predicted WTP at different covariate values from a single pooled regression.

This approach allows the marginal disutility of travel cost (β_1) to differ across demographic groups, and β_1 is the main channel where heterogeneity in WTP per trip can arise for our logit travel cost model. Holding β_1 fixed across groups (e.g., by using a single pooled regression and only varying covariate values) would mechanically average out most of the cross-group variation.

We first modify our preferred travel cost model by dropping the subsetting variable from the right-hand side when it is constant within the group (e.g., race is dropped from the race-specific regressions, rural from the metro-status regressions). For each subgroup, we then run the modified specification of Equation 1 on its observations. Within each subgroup, the average WTP per trip is computed as in Section 3 (Equation 2) and standard errors are computed by the same delta-method procedure, using the subgroup's coefficient covariance matrix.

We estimate the travel cost model two different ways for each of the four demographic groups (race/ethnicity, age group, metro status, and income quintile); first, pricing travel time at one-third of the respondent's average hourly wage (the preferred specification) and second, pricing it at the replacement wage of \$16.08 per hour. Income quintiles are constructed from the empirical distribution of family income in real 2023 USD. Q1 is the bottom 20%, and Q5 is the top 20%.

Figure 3 shows that the gap between the two pricing methods is largest for groups whose wages differ most from the replacement wage. The highest-income quintile (Q5) and White respondents both show much higher WTP under the one-third-wage price than under the flat replacement wage.

C Matching to Infer Trip Behavior for Non-Recreators

This section details how we impute counterfactual travel times for respondents who did not take a recreation trip and how the preferred matching specification was selected, shows how the consumer surplus estimates reported in the main text respond to the number of matched recreators and the caliper, and reports a placebo test of the imputed travel times.

C1 Matching Procedure

We only observe travel time for respondents whose diary day includes a local outdoor recreation trip. For non-recreators, we use nearest-neighbor propensity score matching to impute the travel time they would have faced had they taken a trip, and then use that imputed travel time to calculate their travel cost π_i. We pair every non-recreator with recreators drawn from the same U.S. state and the same three-year period. Each match therefore holds fixed the state-level supply of recreation opportunities and the broader time period. Within-state matching is complementary to the state fixed effects in Equation 1 because matching ensures that imputed travel times come from geographically comparable recreators, while the state fixed effects control for state-level heterogeneity in the likelihood of taking a recreation trip in the travel cost model itself.

Within these state-by-period subsets, we estimate a propensity score $P_i = \Pr(\text{no trip}_i = 1 \mid \mathbf{Z}_i)$ (*i.e.,* the predicted probability that respondent i is a non-recreator) using a logistic regression. Non-recreators are coded as the "treated" group (no trip = 1) and recreators as controls (no trip = 0), so that matching pairs each non-recreator with similar recreators from whom travel time can be imputed. The log-odds of being a non-recreator are modeled as

$$\log\left(\frac{P_i}{1-P_i}\right) = \alpha_1 H_i^S + \alpha_2 H_i^W + \alpha_3 H_i^E + \alpha_4 B_i + \alpha_5 \text{Hol}_i + \alpha_6 R_i + \alpha_7 \text{Age}_i + \alpha_8 \text{Male}_i \quad (6)$$

where H^S, H^W, and H^E are the minutes the respondent spent sleeping, working, and eating or socializing on the diary day, B and Hol indicate whether the diary day fell on a weekend or holiday, R indicates non-metropolitan residence, and Age and Male are the respondent's age and sex. The time-use and calendar variables describe the time constraints of the diary day, so matching on them pairs each non-recreator with recreators who had similar time available for a trip. Residence, age, and sex further pair non-recreators with recreators who live in similar settings and are at a similar stage of life. Combined with matching within state and period, this means matched recreators face a comparable set of recreation opportunities in both space and time. Every variable in Equation 6 constrains or predates the decision to

take a trip. Residence, age, and sex also enter the travel cost model as controls; because they are determined before the trip decision, matching on them sharpens the comparison without conditioning on consequences of the choice being modeled. For the same reason, we exclude same-day car travel and childcare from the preferred equation, since both may result from the decision to take a trip.

We compare Equation 6 with five alternative propensity-score specifications. Three are sparser: time use alone; time use with the weekend and holiday indicators; and time use, calendar indicators, and non-metropolitan residence. Two are richer and progressively add an indicator for traveling by car on the diary day (Car) and an indicator for performing household childcare activities (CC).

We implement nearest-neighbor matching with replacement on $\hat{P}_i$ using the `MatchIt` R package (Greifer, 2025). Each non-recreator's imputed travel time is the simple average of her n matched neighbors' observed travel times. We compare balance across all six propensity-score equations at $n = 3$ and $n = 20$, and, for the preferred equation, estimate the travel cost model for every n from 1 to 25.

We impose caliper restrictions to enforce match quality. For each propensity-score equation and each n, we re-run matching at calipers of 0.05, 0.25, and 1.0, retaining only matches whose propensity scores differ by less than the caliper, measured in standard deviations of the propensity score. Tighter calipers exclude poor matches but discard non-recreators with no sufficiently similar neighbor. Non-recreators left unmatched are dropped from the travel cost model.

We select our preferred matching specification with two goals in mind. First, we want imputed travel times that are as close as possible to the travel times non-recreators would have actually faced had they taken a trip. We use covariate balance as our diagnostic, computing the absolute standardized mean difference (|SMD|) for each covariate before and after matching, and then averaging the post-matching |SMD| across state-by-period subgroups. We assume that matched recreators with better balance experience travel times for recreation that serve as a good proxy for the travel times their matched non-recreators would have faced. Second, we want WTP estimates that do not hinge on an arbitrary choice of how many neighbors to average. We therefore choose the propensity-score equation and caliper on balance alone, before estimating any travel cost model, and then choose n from the range over which WTP per trip is stable (Subsection C3).

For non-recreators, we construct the trip-quality vector $\mathbf{D}_i$ in Equation 1 from the same matched neighbors. Recreators are assigned a one for their observed outdoor activity category and zeros elsewhere. Non-recreators are assigned the share of their n matched neighbors who participated in each category. The resulting $\mathbf{D}_i$ summarizes the expected

composition of activities a non-recreator would most likely undertake had she taken a trip.

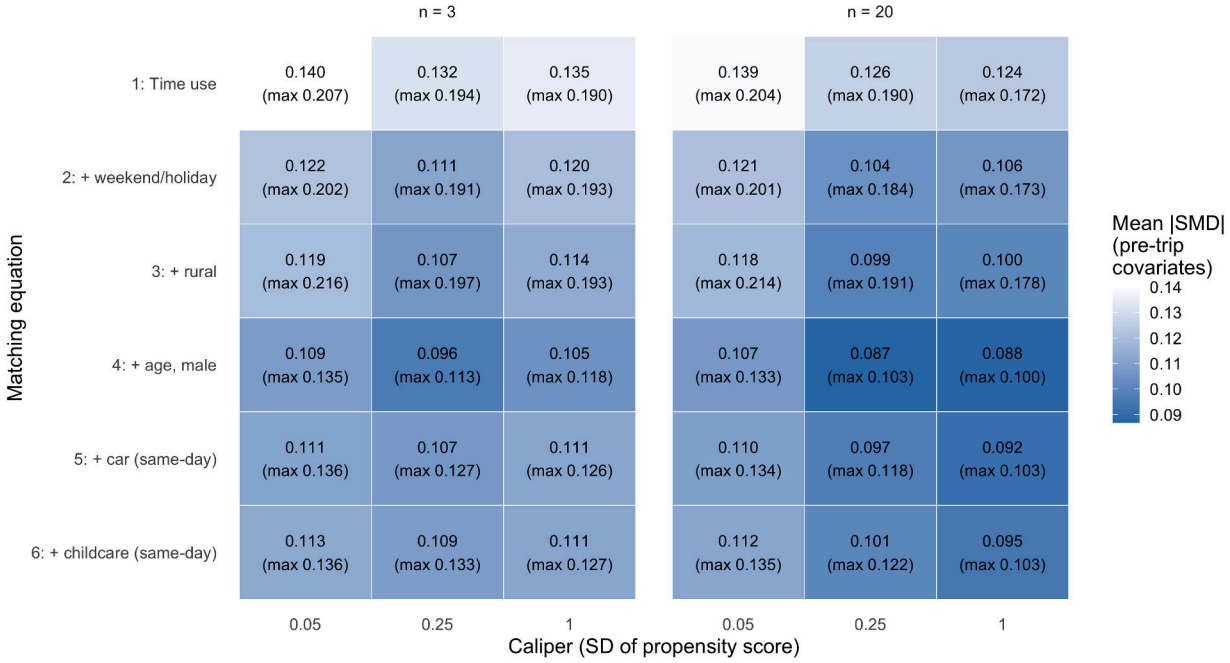
C2 Selecting the Preferred Matching Specification

The matching procedure in Subsection C1 produces matched datasets that vary along three dimensions: the propensity-score equation (which variables enter P_i: six options ranging from the time-use baseline to the full specification with calendar, geographic, demographic, car, and childcare controls), the number of matched recreators per non-recreator (n), and the caliper ($c \in \{0.05, 0.25, 1.0\}$).

We first choose the propensity-score equation and caliper on covariate balance. For every state-by-period subset, propensity-score equation, caliper, and $n \in \{3, 20\}$, we compute the absolute standardized mean difference (|SMD|) between non-recreators and their matched recreators on each covariate after matching, then average across covariates and state-by-period subsets to obtain a single balance score. We score every equation on the same set of pre-trip covariates: the time-use and calendar variables, non-metropolitan residence, age, and sex. A common set is necessary because matching balances the covariates in its own equation; scoring each equation on its own covariates would favor richer equations mechanically. Same-day car travel and childcare are excluded from the common set because they may be outcomes of the trip decision. Figure C1 reports the mean and maximum post-matching |SMD| across these covariates for every combination of propensity-score equation and caliper, at $n = 3$ and $n = 20$.

Based on these diagnostics, we select the propensity-score equation in Equation 6, which adds non-metropolitan residence, age, and sex to the time-use and calendar variables. It achieves the lowest mean |SMD| of the six equations at every caliper and both values of n (Figure C1). With this equation, a caliper of 0.25 balances better than 1.0 at $n = 3$ (mean |SMD| of 0.096 versus 0.105), and the two are equivalent at $n = 20$ (0.087 versus 0.088), where no covariate exceeds an |SMD| of 0.103 under a caliper of 0.25. A caliper of 0.05 balances worse at both values of n and leaves 16 percent of non-recreators unmatched, compared with 3 percent at 0.25. We therefore select a caliper of 0.25. The next subsection chooses n and shows how WTP per trip responds to n under each caliper.

Figure C1: Post-Matching Covariate Balance by Matching Specification



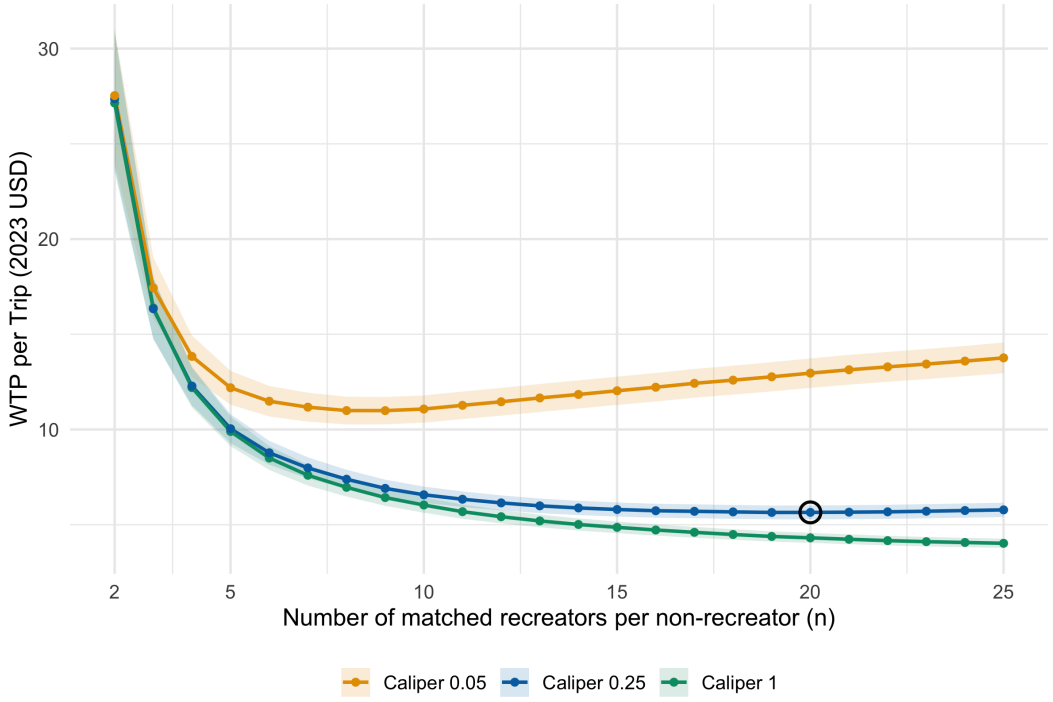
Post-matching balance between non-recreators and their matched recreators for each propensity-score equation and caliper, at $n = 3$ and $n = 20$ matched neighbors. Each cell reports the mean absolute standardized mean difference (|SMD|) and, in parentheses, the largest |SMD| across a common set of pre-trip covariates: minutes sleeping, working, and eating or socializing; weekend and holiday indicators; rural; age; and sex. Same-day car travel and childcare are excluded from the common set because they may be outcomes of the trip decision. Darker shading indicates better balance.

C3 Robustness Across Matching Specifications

Figure C2 plots pooled WTP per trip against the number of matched recreators per non-recreator, n, for the preferred propensity-score equation under each caliper. With few neighbors, WTP per trip is high and imprecise: at $n = 2$, it is about \$27 under every caliper. A non-recreator's imputed travel cost is then an average over only a few recreators' trips and carries substantial idiosyncratic noise. Noise in a regressor attenuates its coefficient toward zero, so it biases the travel-cost coefficient $\hat{\beta}_1$ toward zero and inflates WTP. At $n = 1$, $\hat{\beta}_1$ is close to zero and WTP is not meaningfully estimated, so the figure begins at $n = 2$. As n rises, averaging over more neighbors removes this noise, $\hat{\beta}_1$ grows in magnitude, and WTP per trip falls.

Under a caliper of 0.25, WTP per trip levels off: it ranges only from \$5.65 to \$5.80 for n between 15 and 25, a spread smaller than one standard error (\$0.18–\$0.19). We therefore select $n = 20$, which lies in this flat region, so the preferred estimate does not depend on the precise number of neighbors. The other two calipers do not level off. Under a caliper

Figure C2: WTP per Trip by Number of Matched Recreators and Caliper



Average WTP per local recreation trip (all years pooled, preferred travel cost model M11, travel time priced at one-third of the wage) as the number of matched recreators per non-recreator, n, varies from 2 to 25. Each line is one caliper; all use the preferred propensity-score equation. Shaded bands are 95% confidence intervals from delta-method standard errors. The circled point is the preferred specification ($n = 20$, caliper $= 0.25$). Real 2023 USD.

of 1.0, WTP per trip continues to fall through $n = 25$ ($4.03), consistent with a wide caliper allowing large n to draw increasingly dissimilar recreators and pulling imputed travel times toward the state-by-period average. Under a caliper of 0.05, WTP per trip reaches a minimum of about $11 near $n = 8$ and then rises to $13.76 at $n = 25$. Only the 0.25 caliper both balances as well as or better than the alternatives and yields a WTP per trip that is stable in n.

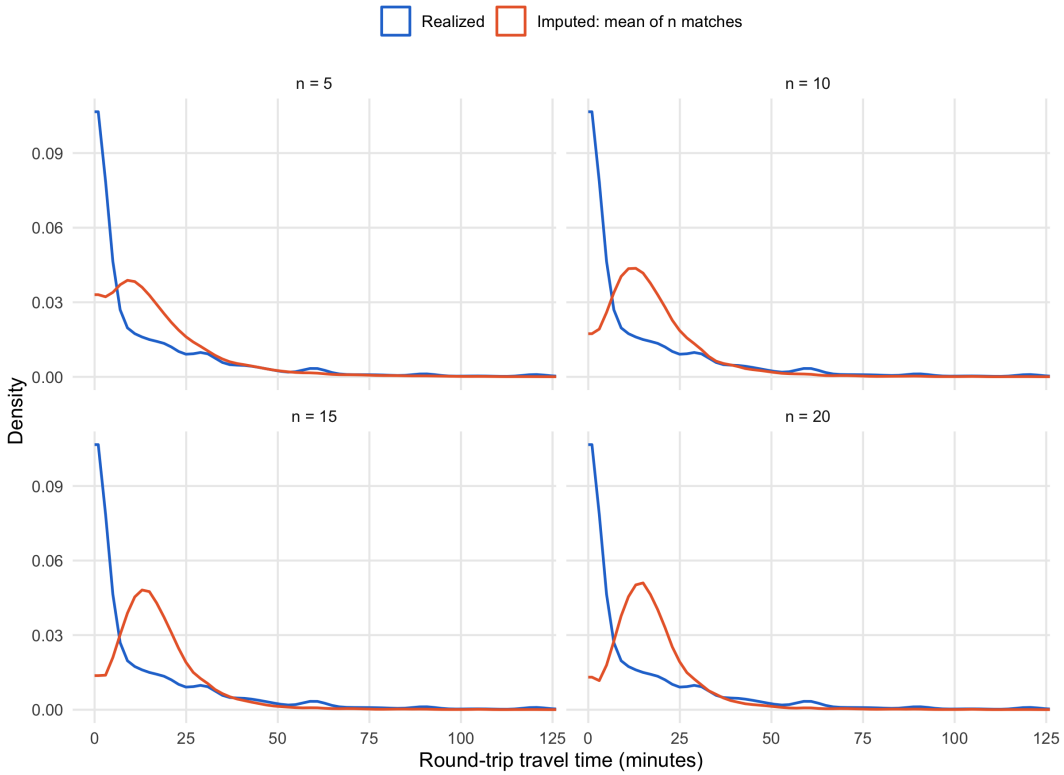
C4 Placebo Test of Imputed Travel Times

To assess how well matching recovers travel time, we apply the imputation to recreators, whose travel times we observe. Within each state-by-period subset, we randomly divide recreators into five groups. We treat each group in turn as if its members were non-recreators and match them to the recreators in the other four groups, using the propensity score from the main matching procedure (Equation 6) and a caliper of 0.25. As for non-recreators, each person's imputed travel time is the average over her n matched neighbors. Every recreator

therefore receives one imputed travel time, which we compare with her observed travel time. At least one match within the caliper is found for 97 percent of recreation trips (35,037 of 36,159).

Figure C3 compares the distributions of observed and imputed travel times for $n \in \{5, 10, 15, 20\}$. Matching recovers average travel time: at $n = 20$, the mean imputed travel time is 18.4 minutes, compared with an observed mean of 18.1 minutes, and the mean is essentially unchanged across n. The shapes of the two distributions differ. Observed travel times are concentrated at zero: 41 percent of recreation trips involve no travel time. Each imputed travel time is an average over n neighbors, so imputed values are pulled toward the middle of the distribution. At $n = 20$, 2 percent of imputed travel times are zero, and the median imputed travel time is 15.8 minutes, compared with an observed median of 6 minutes. Larger values of n concentrate the imputed distribution further around 15 minutes. This difference in shape is an artifact of imputing, rather than observing, travel time for non-recreators.

Figure C3: Observed and Imputed Travel Times for Recreators (Placebo Test)



Density of round-trip travel time for recreation trips. "Realized" is the travel time observed in the time diary. "Imputed" is the travel time assigned when the recreator is treated as a non-recreator and matched to n recreators in the same state and three-year period (preferred propensity-score equation, caliper 0.25), computed as the average of the matched recreators' travel times. Each panel is one value of n. Includes all recreation trips with at least one match (35,037 trips); the horizontal axis is truncated at 120 minutes.

D Natural Capital Stock Changes and the Relative Price Change Framework

This section tests and inverts the relative price change framework of Drupp et al. (2024, 2025) using our 20-year series of per-trip WTP. It states the test and back-out procedures, reports the implied growth of recreation-enabling assets, and assesses robustness to the measure of consumption growth and to the assumed benchmark for asset growth.

D1 Inferring Natural Capital Stock Changes

Our estimates of per-trip WTP for local outdoor recreation in real 2023 USD give us a 20-year repeated measure of the unit price of an ecosystem service. This series allows us to empirically test the relative price change (RPC) framework proposed by Drupp et al. (2024)

for how the value of an ecosystem service evolves. The relative price change framework, restated in Equation 5 of the main text, predicts that the real relative price of an ecosystem service changes according to $\text{RPC}_t = \xi\,(g_{C,t} - g_{E,t})$, where ξ is the income elasticity of WTP for the service, $g_{C,t}$ is the growth rate of real consumption per capita, and $g_{E,t}$ is the growth rate of the underlying ecosystem service flow. We use per-trip WTP as the relevant unit price. Our aggregated values (annual per-capita value and national value) reflect growth in the number of trips taken, driven by population growth and rising participation. One might mistakenly interpret rising total or per-capita value as a scarcity signal for the underlying asset. Because these aggregated values rely on changes in quantity demanded, they cannot serve as signals of scarcity of the underlying asset—the per-trip willingness to pay reflects how the marginal value of a single trip has changed.

We use the RPC equation (Equation 5) in two complementary ways. The first is a forward test of the framework. Taking the observed path of per-capita consumption growth $\{g_{C,\tau}\}$ and an assumed value of g_E, we project the per-trip WTP path that the framework predicts by compounding the implied RPCs from the first observed period forward,

$$\widehat{\text{WTP}}_t^{\text{Drupp}} \;=\; \text{WTP}_0 \cdot \prod_{\tau=1}^{t} \big(1 + \xi\,(g_{C,\tau} - g_{E,\tau})\big), \tag{7}$$

and compare $\widehat{\text{WTP}}_t^{\text{Drupp}}$ to our directly estimated $\overline{\text{WTP}}_t$. The framework passes the test for this service if the projected path lies within the confidence intervals of the estimated path. The second is to back out g_E by rearranging the RPC equation and using RPC_t recovered from our estimated WTP series,

$$g_{E,t} \;=\; g_{C,t} - \text{RPC}_t/\xi, \tag{8}$$

where

$$\text{RPC}_t \;=\; \frac{\overline{\text{WTP}}_t - \overline{\text{WTP}}_{t-1}}{\overline{\text{WTP}}_{t-1}}. \tag{9}$$

When the relative price change framework is taken as given, Equation 8 yields the implied growth rate of the recreation-enabling asset stock. This addresses a key challenge in natural capital accounting when the underlying assets (parks, greenways, tree-lined streets, mixed-use spaces) are too dispersed to be inventoried directly.

We do not estimate ξ from our data. Instead, we use four benchmark values from the meta-analysis in Drupp et al. (2025) as a sensitivity range: $\xi = 0.60$ (the central global estimate for ecosystem services), $\xi = 0.74$ (recreation and ecotourism), $\xi = 0.86$ (North

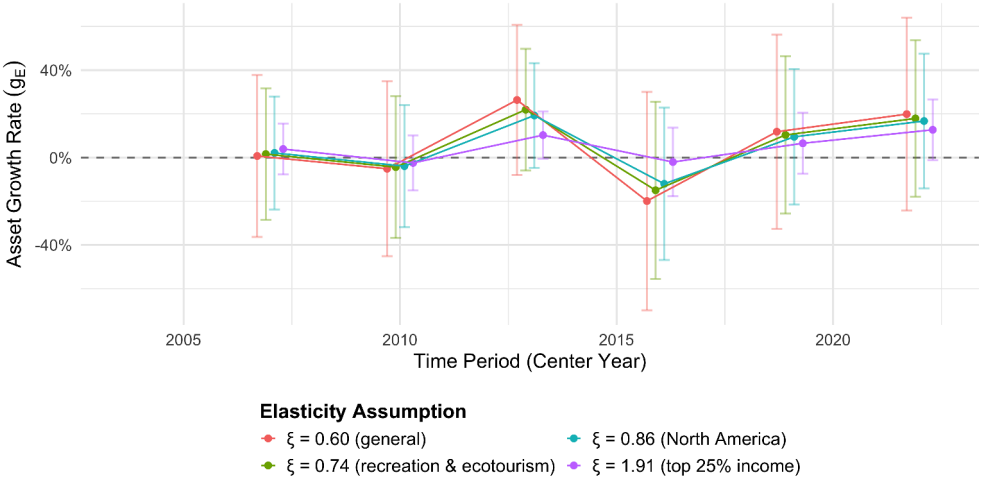
American studies), and $\xi = 1.91$ (the top-income-quartile estimate).

Our preferred measure of the growth rate of consumption, $g_{C,t}$, is real personal consumption expenditures per capita (FRED series `A794RX0Q048SBEA`). We aggregate real personal consumption expenditures per capita to the same three-year periods as our WTP estimates. Expenditure per capita is the theoretically appropriate measure because the RPC framework follows the Ramsey consumption-discounting tradition in which the relevant marginal-utility argument is consumption, not income or output (Ramsey, 1928; Sterner and Persson, 2008; Groom and Maddison, 2019). As a robustness check, we re-run all analyses with real GDP per capita (FRED series `A939RX0Q048SBEA`) as g_C, which is the proxy most commonly used in the natural-capital-accounting literature (Drupp et al., 2024, 2025; Addicott et al., 2020). Results under GDP per capita are reported in Subsection D3.

D2 Implied Asset Growth g_E (Per Capita Expenditure as g_C)

Section 5 reports the average implied g_E from the back-out exercise. Figure D1 and Table D1 below show the period-by-period values that those averages are computed from, using real personal consumption expenditures per capita as g_C.

Figure D1: Implied Growth Rate of Recreation-Enabling Assets (g_E)



Implied growth rate of recreation-enabling natural capital g_E for each three-year period, computed by solving Equation 8 using observed per-trip WTP and real per capita expenditure as g_C. Four ξ benchmarks from Drupp et al. (2025) are shown. Error bars are 95% confidence intervals propagated from delta-method WTP standard errors. Travel time is priced at one-third of the respondent's average hourly wage.

Table D1: Implied g_E by Three-Year Period (Expenditure per Capita as g_C)

Period	RPC_t	g_C (Expend. p.c.)	g_E (ξ=0.60)	g_E (ξ=0.86)	g_E (ξ=0.74)	g_E (ξ=1.91)
2003–2005	—	—	—	—	—	—
2006–2008	2.8%	5.4%	0.7%	2.1%	1.6%	3.9%
2009–2011	2.3%	−1.2%	−5.1%	−4.0%	−4.4%	−2.5%
2012–2014	−14.1%	2.9%	26.3%	19.3%	21.9%	10.3%
2015–2017	15.6%	6.1%	−19.9%	−12.0%	−15.0%	−2.1%
2018–2020	−4.6%	4.1%	11.8%	9.5%	10.4%	6.5%
2021–2023	−6.3%	9.4%	19.9%	16.7%	17.9%	12.7%
Average	$\approx -1\%$	$\approx 4\%$	$\approx 5.6\%$	$\approx 5.3\%$	$\approx 5.4\%$	$\approx 4.8\%$

RPC_t = period-over-period change in per-trip WTP (real 2023 USD). g_C = change in real personal consumption expenditures per capita (FRED `A794RX0Q048SBEA`). g_E is computed by solving Equation 8. The large, offsetting changes in per-trip WTP in 2012–2014 and 2015–2017 produce correspondingly large swings in g_E in those periods. All values in percent.

D3 Robustness: GDP per Capita as g_C

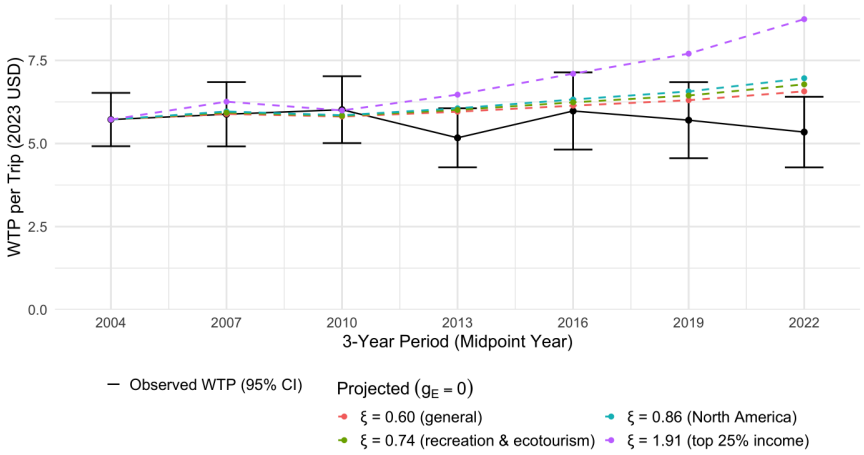
The main-text RPC results use real personal consumption expenditures per capita as g_C, for the reasons given in Subsection D1. Most empirical work, however, proxies g_C with real GDP per capita because per capita consumption data are unavailable for most countries and periods (Drupp et al., 2024, 2025; Addicott et al., 2020). For comparability with that literature, we repeat both exercises (the test and the back-out) using real U.S. GDP per capita (FRED `A939RX0Q048SBEA`) as g_C.

Over our study period, real GDP per capita grew slightly more slowly than real consumption expenditures per capita (32% versus 36% from 2003 to 2023), so the two measures yield very similar results. Figure D2 repeats the test of Section 5 using GDP per capita. As under PCE, the projected paths for $\xi \in \{0.60, 0.74, 0.86\}$ lie within the 95% confidence intervals of the observed WTP series in every period except 2021–2023, and the $\xi = 1.91$ projection falls outside them in 2012–2014, 2018–2020, and 2021–2023. The implied g_E values (Table D2) are correspondingly close to those under PCE. Average implied g_E ranges from +4.3% ($\xi = 1.91$) to +5.1% ($\xi = 0.60$) per three-year period, compared with +4.8% to +5.6% under PCE. The conclusions of Section 5 therefore do not depend on the choice of consumption measure.

D4 Sensitivity to the Assumed g_E in the Test

The main-text test fixes $g_E = 0$, the baseline of an unchanged asset stock (Section 5). Figure D3 compares this baseline with two negative asset growth rates drawn from Drupp et al. (2025): $g_E = -1.01\%$ (their aggregate ecosystem service decline) and $g_E = -0.11\%$

Figure D2: Test of the RPC Framework Using GDP per Capita as g_C



Observed per-trip WTP (black, with 95% CI error bars) vs. projected path from Equation 7 compounded forward from 2003–2005 under $g_E = 0$. Colored lines correspond to four income elasticity benchmarks from Drupp et al. (2025). g_C is real U.S. GDP per capita. All values in real 2023 USD.

Table D2: Implied g_E by Three-Year Period (GDP per Capita as g_C)

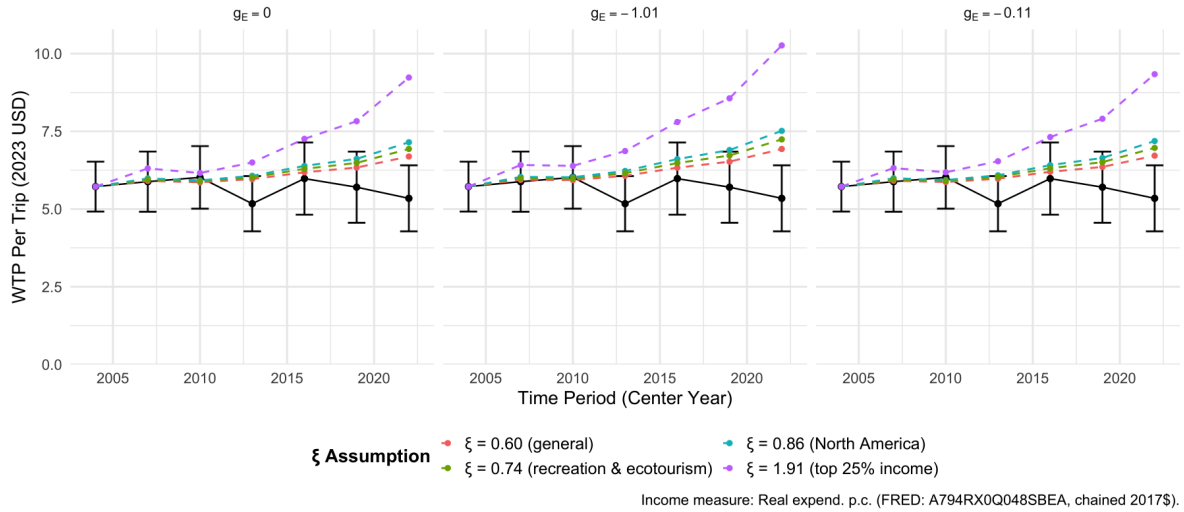
Period	RPC$_t$	g_C (GDP p.c.)	g_E (ξ=0.60)	g_E (ξ=0.86)	g_E (ξ=0.74)	g_E (ξ=1.91)
2003–2005	—	—	—	—	—	—
2006–2008	2.8%	4.9%	0.3%	1.7%	1.1%	3.5%
2009–2011	2.3%	−2.2%	−6.1%	−4.9%	−5.4%	−3.4%
2012–2014	−14.1%	4.2%	27.6%	20.5%	23.2%	11.5%
2015–2017	15.6%	5.1%	−20.9%	−13.1%	−16.0%	−3.1%
2018–2020	−4.6%	4.4%	12.2%	9.8%	10.7%	6.9%
2021–2023	−6.3%	7.0%	17.5%	14.4%	15.5%	10.3%
Average	$\approx -1\%$	$\approx 4\%$	$\approx 5.1\%$	$\approx 4.7\%$	$\approx 4.9\%$	$\approx 4.3\%$

RPC$_t$ = period-over-period change in per-trip WTP (real 2023 USD). g_C = change in real U.S. GDP per capita (FRED `A939RX0Q048SBEA`). g_E is computed by solving Equation 8. All values in percent.

(global forest area decline). Because g_C and the RPC are measured as growth per three-year period, all assumed values of g_E in this subsection are applied per three-year period; for the two Drupp et al. (2025) values, which are annual rates, this implies a slower decline than they report. Negative values of g_E raise the projected path and so widen its gap with our estimates. Under $g_E = -0.11\%$, the periods in which each projection falls outside the confidence intervals are the same as under $g_E = 0$. Under $g_E = -1.01\%$, the projections for $\xi \in \{0.60, 0.74\}$ also fall outside in 2012–2014, and the projection for $\xi = 0.86$ also falls outside in 2018–2020. Figure D4 reports the converse calculation at the recreation-and-ecotourism elasticity ($\xi = 0.74$). We sweep g_E over a fine grid and plot the projected WTP

paths at the edges of the set of values for which the projection remains inside the observed 95% confidence intervals in every period. Under this elasticity, the framework's projection stays within our confidence intervals only for g_E between +1.9% and +8.9% per three-year period.

Figure D3: Test Sensitivity to Assumed g_E

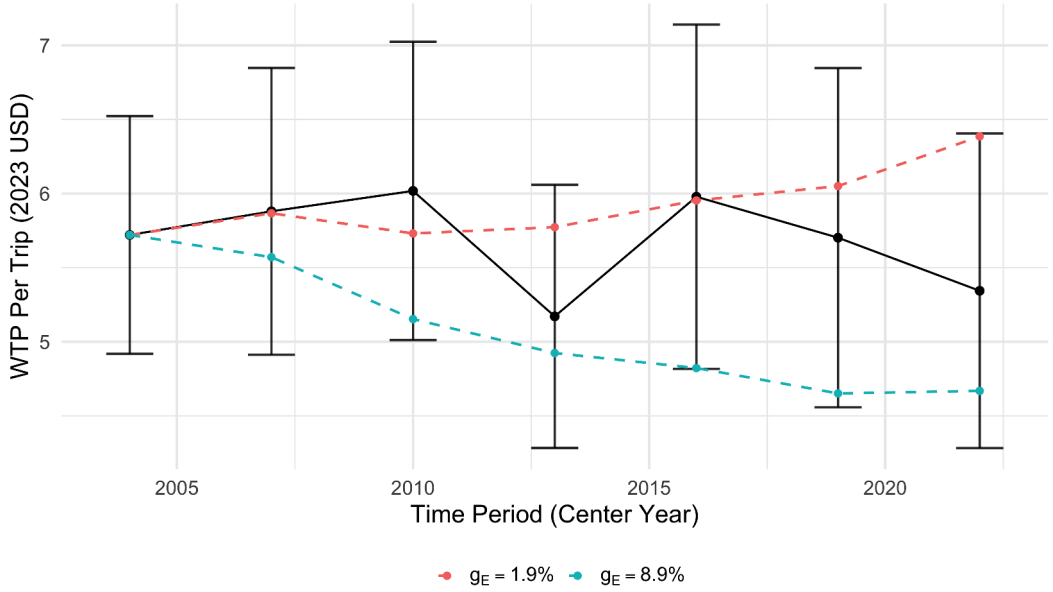


Observed per-trip WTP vs. projected WTP from Equation 7 under three assumed asset growth rates: $g_E = 0$ (stock maintained), $g_E = -0.11\%$ (global forest decline), and $g_E = -1.01\%$ (aggregate ecosystem service decline) from Drupp et al. (2025). g_C is real PCE per capita.

D5 Supply and Demand Benchmarks for g_E

Two independent data sources indicate that, relative to the $g_E = 0$ baseline in Section 5, the recreation-enabling assets have, if anything, grown. Table D3 reports the supply-side growth: total park acreage across the 100 largest U.S. cities grew at roughly 9.8% per three-year period from 2010 to 2022 (The Trust for Public Land, 2025). Table D4 reports the demand-side growth: total national local outdoor recreation trips from ATUS grew on average 10.7% per three-year period from 2003 to 2023.

Figure D4: Range of Asset Growth Rates Consistent with the RPC Framework



Observed per-trip WTP (black, with 95% CI error bars) vs. projected WTP from Equation 7 at the recreation-and-ecotourism income elasticity ($\xi = 0.74$), evaluated at the edges of the set of assumed asset growth rates for which the projection remains inside the observed confidence intervals in every period: $g_E = +1.9\%$ and $g_E = +8.9\%$ per three-year period. g_C is real PCE per capita.

Table D3: Park Acreage Growth by Three-Year Period (TPL, Top 100 U.S. Cities)

Period	Avg. Park Acres (Top 100 Cities)	g_E (Acreage)
2009–2011	1,457,649	–
2012–2014	1,567,312	7.5%
2015–2017	1,970,106	25.7%
2018–2020	2,098,322	6.5%
2021–2023	2,087,218	-0.5%
Average		$\approx$ 9.8%

Average total park acres across the 100 largest U.S. cities per three-year period. Source: Trust for Public Land City Park Facts.

Table D4: Total Local Recreation Trips by Three-Year Period

Period	Total Trips (Millions)	Period Growth
2003–2005	9,964	–
2006–2008	9,230	-7.4%
2009–2011	11,839	28.3%
2012–2014	11,983	1.2%
2015–2017	14,279	19.2%
2018–2020	13,770	-3.6%
2021–2023	17,437	26.6%
Average		$\approx 10.7\%$

Average annual national local outdoor recreation trips per three-year period (ATUS). Period growth is the change from the previous three-year period.